

Displacement or Augmentation? The Effects of AI Innovation on Workforce Dynamics and Firm Value

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Abstract: We apply large language models (LLMs) and generative AI to U.S. patent data during 2007-2023 to identify and classify AI-related innovations in seven key functional areas. Relying on exogenous variation in AI patenting tied to the leniency of randomly-assigned patent examiners, we identify the causal effects of firm-level Artificial Intelligence (AI) innovation on employment dynamics and corporate valuation. Using microdata on individual worker skill sets and job transitions between occupations and firms, we find that AI innovations related to high-level cognition (i.e., language, decision-making, engagement, learning, or creativity) augment human labor, while those related to low-level cognition (i.e., perception or motor control) displace it. Among augmenting AI innovations, those with innovative cognitive styles (i.e., engagement, learning, or creativity) tend to increase demand for labor with novel skills. Although both augmenting AI innovations and displacing AI innovations significantly increase firm value, they appear to do so through different channels. In particular, augmenting innovations raise firm-level productivity, whereas displacing innovations improve cost efficiency. We also find that firms obtain more value from their augmenting (displacing) innovations when access to external hires is better (potential cost savings from workforce reduction are larger). Overall, our findings suggest that AI innovation can bring significant value gains to innovating firms, but the extent of those gains depends critically on what frictions are present in external labor markets.

JEL Classification Codes: G30, G32, J23, J24, O32, O33

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1. Introduction

The rapid development of artificial intelligence (AI) technologies over the past few years has led to growing interest among academics, practitioners, and policymakers in understanding the possible effects of systems that can process vast amounts of information, generate predictions and original content, and engage with human users. Many view AI technologies as a key driver of future economic growth (Aghion et al., 2017; Agrawal et al., 2019a), with a large part of the gains coming from AI's ability to complement and augment the productivity of knowledge workers.³ At the same time, questions have arisen about the implications of AI technology for the labor market. As increasingly powerful AI systems begin to perform cognitive tasks that have traditionally been within the domain of skilled human labor, concerns have grown that widespread use of AI may lead to the displacement of skilled knowledge workers (Frey and Osborne, 2017), who have traditionally been considered as the beneficiaries of the spread of older automation technologies.⁴

Does AI technology complement human employees in cognitive tasks, thereby augmenting workforce productivity and creating new jobs and new demand for skilled labor?⁵ Or is it the case that AI can perform workplace tasks more inexpensively and more efficiently than humans can, leading to skill obsolescence, labor displacement, and higher unemployment in the future? Despite the obvious economic and public-policy importance of such questions, they have been difficult to address due to the empirical challenge of quantifying firm-level AI adoption and the dynamics of AI exposed jobs (Seamans and Raj, 2018). Further adding to the difficulty is the fact that AI, being a general-purpose technology, includes a remarkably broad range of components, functionalities, and real-world applications (Brynjolfsson and McAfee, 2017; Caliskan and Lum, 2024). Thus, a complete understanding of the labor-market effects of AI requires being able to distinguish among different types of AI that may have very different effects on firms' labor demand.

Recent academic research has focused on the labor effects of either AI technology as a whole or of one specific subset—generative AI—but the empirical evidence remains inconclusive, and

³ McKinsey (2025) estimates that AI could unlock \$4.4 trillion in annual productivity gains globally, with companies using AI showing significant improvements in efficiency.

⁴ Reflecting such concerns, an International Monetary Fund report (Cazzaniga, 2024) concludes that about 30% of jobs in advanced economies are set to be negatively affected by AI, resulting in lower salaries, reduced hiring, and possible job destruction.

⁵ A recent OECD (2024) report emphasizes the importance of training and education to equip both current and future workers with the necessary skills and to support displaced workers with adequate social protection in the coming AI revolution. The report also concludes that a widespread scarcity of specialized skills is hindering AI adoption, highlighting a need for updated education policies to bridge the talent gap. See <https://www.oecd.org/en/topics/policy-issues/artificial-intelligence.html>.

the reasons for the different findings are not well understood. In this paper, we use textual data from millions of patents over 2007-2023 and microdata on worker flows to provide evidence on the causal effects of different AI types on firm-level employment, efficiency, and value. To enable the analysis, we develop a new typology for classifying AI technologies in terms of functional capabilities that parallel major aspects of human cognition. Applying state-of-the-art generative AI and large language models (LLMs) to patent filing texts, we classify AI innovations into seven overlapping functional areas: Language, Perception, Motor Control, Engagement, Decision-making, Learning, and Creativity.⁶

We document a number of key facts about the growth of AI innovation across functional areas, technological areas, and industrial sectors. For instance, we find that, compared to all patenting in the U.S., AI patenting has grown extremely rapidly over the past decade. Although the largest source of the more than 140,000 AI innovations in our sample is firms in the Manufacturing sector, a substantial amount also occurs across other industries such as Transport, Storage, and Communications. The most common types of AI innovation in our sample are decision-making AI and learning AI. Decision-making AI is the single fastest-growing type, likely reflecting the tremendous growth in data analysis, machine-learning algorithms, and generative AI tools in recent years.

In the main part of our analysis, we examine whether different types of AI innovation have different occupation-specific effects on firms' employment. For this purpose, we use a micro-level dataset that includes information on the job transitions of individual workers to, from, and within firms in our sample. The dataset, obtained from Revelio Labs, covers employee data at nearly all U.S. publicly-traded firms (including their subsidiaries) from 2008 to the present. In addition to providing insights into worker movements at a monthly frequency, the dataset also has information on employees' occupations and skill sets, enabling us to capture how firms' net hiring of skills and workers within different occupational segments changes in response to AI innovation.

To identify the extent to which each occupation is exposed to AI, we link the data from O*NET on the skill requirements of different occupations to the functional capabilities under each type of AI technology. By doing so, we distinguish between employment effects that are likely the result of direct complementarities or substitutions between AI systems and human workers versus

⁶ Section 2 discusses in detail the rationale for this typology and provides definitions and examples for each type.

those effects that merely reflect indirect employment effects, between-occupation spillovers, or shifts in the firm’s labor demand due to a re-prioritization of various occupations.

Aside from exploring which types of AI innovation have labor-displacing and/or labor-augmenting effects, a second aim of our study is to shed light on the different ways in which AI can result in workforce augmentation. We posit two different kinds of AI-driven labor augmentation. On the intensive margin, AI innovation may complement existing jobs, enhancing productivity and enabling firms to scale up their core business (Acemoglu and Autor, 2011; Acemoglu et al., 2022b). On the extensive margin, AI innovation may allow firms to expand their business scope by exploring new products, diversifying into new industries, and creating new jobs (Acemoglu and Autor, 2011; Agrawal et al., 2023; Cockburn et al., 2019; Babina et al., 2024), fundamentally altering skill preferences. In this case, the new skill demands might include not only AI-specific technical expertise (Korinek and Stiglitz, 2017; Alekseeva et al., 2021; Acemoglu et al., 2022b), but also skills that complement the new business areas created by AI (Frank et al., 2019; Makarius et al., 2020).⁷ However, acquiring new skills may present significant challenges for firms as they face hiring frictions related to potential skill gaps and mismatches (McGuinness et al., 2018).

Given both the recognized importance of new skills for AI adoption and the substantial costs associated with hiring, it is not clear *a priori* to what extent AI will drive firms to actively invest in acquiring new skills. To empirically examine whether AI increases labor demand by scaling existing roles or by expanding the range of labor inputs, we make use of data on individual workers’ skill sets from Revelio Labs to quantify the changes in employment with (without) novel skills that were absent (present) from a given occupation. For each type of AI innovation, we can then examine whether it induces firms to accumulate workers with new skills in those occupations exposed to the relevant AI type.

In studying the link between AI innovation and firm-level employment, one important issue that arises is potential endogeneity: observed relations between innovation and employment could reflect non-causal correlations arising from the influence of other factors. For example, AI patenting activity might be correlated with changes in labor markets, industries, consumer demand,

⁷ Recent studies have explored firm-level investment in AI-specific skills as evidence for AI adoption (Alekseeva et al., 2021; Babina et al., 2024). However, the effects of firm-level AI adoption on a more general set of novel worker skills within the firm have not been previously examined.

or corporate policies (e.g., overall patenting or R&D spending) that are the true drivers of changes in firms' labor-market strategies. To address these concerns, we follow the approach of prior work that uses the quasi-random assignment of patent applications to patent examiners within different U.S. Patent and Trademark Office (USPTO) art units (Gaule, 2018; Sampat and Williams, 2019; Farre-Mensa, Hegde, and Ljungqvist, 2020). Given that some patent examiners are more lenient than others, the quasi-random assignment of patent applications induces exogenous variation in the probability of AI patent grants within a certain category, which in turn induces variation in the incentives that firms have to adopt and use AI technologies.

We measure employment changes at AI-exposed occupational segments within firms and document in multiple regressions that AI innovation can have both labor-augmenting and labor-displacing effects. In particular, AI innovations related to high-level cognition (i.e., language, decision-making, engagement, learning, or creativity) significantly increase employment in occupations exposed to these types of AI. Innovations related to low-level cognition (i.e., perception or motor control) significantly decrease employment in exposed occupations. Additionally, we find that augmentation can occur in two distinct ways: a net increase in external hires who bring new skills to an occupation ("scope augmentation") and a net increase in overall employment, but not employment of workers who bring new skills ("core augmentation"). Among the augmenting AI types, those corresponding to innovative cognitive styles (i.e., engagement, learning, or creativity) tend to increase demand for labor with novel skills. These results are robust to the use of IV-2SLS (Two-Stage Least Squares) regressions as well as firm fixed effects, occupation-by-year fixed effects, industry-by-year fixed effects, application count-by-year fixed effects, and a variety of time-varying firm-level controls.

Next, to study how AI innovation affects firm-level outcomes, we group the various types of AI patents into four broad categories based on the empirical findings from occupation-level regressions. We define a patent to be *augmenting* if it belongs to any of the AI types—creativity, engagement, learning, language, or decision-making—that are observed to increase overall employment within a firm's occupational segment. A patent is defined as *scope-augmenting* if it is a type of AI that increases employment of workers who bring new skills to a firm's occupational segment: engagement, learning, or creativity. A patent is *core-augmenting* if it is augmenting but not scope-augmenting. Finally, a patent is *displacing* if it is of a type that leads to a reduction in occupational employment, i.e., perception and motor control.

We use regression analysis to explore the causal effects of these broad groups of AI innovation on innovating firms' productivity, costs, and value. A priori, there are several reasons to expect that both augmenting and displacing AI patents can yield benefits to innovators. For instance, new AI tools that enhance the productivity of existing workers in a firm could prompt the firm to scale up production by hiring more workers of the same type. Alternatively, an AI innovation could create value by fostering entirely new jobs or tasks (Bessen, 2018; Acemoglu et al., 2022b) that themselves require additional hiring or reskilling of workers. AI could also lead to increased firm growth and product innovation (Babina et al., 2024). In the case of labor-displacing AI, a firm could potentially reap large cost savings by laying off human workers and substituting for them with automated AI that is less expensive, but equally capable (Seamans and Raj, 2018). We find evidence from IV-2SLS regressions that both augmenting and displacing AI patents yield positive benefits to innovating firms, but they do so through different channels. In particular, augmenting innovations increase total factor productivity (TFP) for the innovating firm. Displacing innovations, however, do not appear to lead to productivity gains for the firm. Instead, displacing AI significantly reduces the firm's SG&A and operating expenses. With regards to corporate valuation, all four broad categories of AI innovation lead to significant increases in innovating firms' Tobin's Q in the year following the patenting activity.

In additional analyses, we explore the circumstances under which labor-augmenting or labor-displacing AI innovation leads to larger value gains. We conjecture that augmenting patents contribute the most to firm value when the frictions of searching, recruiting, and hiring from the external labor market are low. For displacing AI, firms should derive more value when potential cost-saving benefits are large. To examine these predictions, we estimate regressions of firm value on augmenting and displacing AI innovation using subsamples based on cross-sectional differences in characteristics such as labor mobility, enforceability of noncompete agreements (NCAs), unionization rates, unemployment insurance (UI) benefits, and the supply of diverse skills. In general, we find that when firms face greater frictions and impediments to external hiring, the value gains from augmenting AI innovations are lower or insignificant. For labor-displacing AI, value gains are higher with lower UI benefits or a larger quantity of displaceable employees and wages. These findings are broadly consistent with the view that labor-market frictions matter in important ways for both innovation-induced external hiring and worker terminations.

Our paper relates to a large body of work that studies the labor-market implications of technological progress. Papers in this literature show that technologies such as automation or digital computing can substitute for human workers, create new jobs, and lead to skill-biased effects that complement high-skilled labor and displace low-skilled labor (Acemoglu and Restrepo, 2019; Krusell et al., 2000; Acemoglu and Autor, 2011). We extend this literature by using micro-level employment data to study the labor-market effects of AI, a fundamentally important technological development that promises to change how and whether many cognitive tasks in the modern firm are performed by humans. By accounting for individual workers’ occupational roles and their movements between firms, we can provide early causal evidence on AI’s employment effects and help shed light on whether AI in fact complements or substitutes for human labor.

We also contribute to the literature that focuses specifically on the labor-market effects of AI. Some studies find evidence that employment and wages decline after the mainstream arrival of generative AI (Hui et al., 2024; Eisefeldt et al., 2025). Other studies find a positive association between employment growth and the rise of AI technologies (Alderucci et al., 2020; Yang, 2022; Agrawal et al., 2023; Noy and Zhang, 2023; Babina et al., 2024; Eloundou et al., 2024; Autor, 2024). Also, the effects of AI on firms’ labor demand can depend on occupation-specific task exposures to AI (Hampole et al., 2025), and AI can itself alter the skill- or task-based requirements of occupations (Tambe, 2014; Alekseeva et al., 2021; Acemoglu et al., 2022b; Babina et al., 2023). To date, however, researchers have not systematically distinguished among different AI types and the different impacts that they have on worker flows between occupations and firms. Through our use of worker-firm matched data, individuals’ skill sets, patent text data, and generative AI techniques, we are able to fill this gap by quantifying the heterogeneous employment effects of seven different AI types. Overall, our findings on which AI innovation types are labor-augmenting and which are labor-displacing add a new perspective to the broader discourse on AI and provide possible explanations for the mixed evidence in the literature.

Our paper also contributes to a stream of research that studies AI’s effects on other corporate outcomes besides employment growth, such as production efficiency, cost savings, and market value.⁸ Several papers exploit the public release of generative AI tools (e.g., ChatGPT) or

⁸ A related literature examines the implications of AI for various aspects of financial markets, such as price discovery (Lopez-Lira and Tang, 2023), financial analysts (Cao et al., 2024), robo-advising (D’Acunto et al., 2019), asset management (Bartram et al., 2021), and auditing (Fedyk et al., 2022).

prohibitions against their use to study the impacts of the new technology (Kreitmeir and Raschky, 2023; Bertomeu et al., 2023; Hui et al., 2024; Eisfeldt et al., 2025). However, since generative AI is by no means the only type of AI that can have important economic effects on firms and workers, we employ a different approach, relying on the rich and varied data in patent filings to study the effects of a broad range of AI types. Some other studies use survey data to study the real adoption of AI technology and its subsequent effects (e.g., Chatterjee et al., 2021, Rammer et al., 2022). Unlike these studies, our work focuses on actual AI innovation by firms (and their resulting IP rights), and hence we are able to provide causal evidence on the firm-level consequences of direct economic incentives to adopt and use AI technologies.⁹

In a line of research related to ours, Babina et al. (2024) use micro-level data on workers' AI skills to document a positive effect of firms' AI investment on their growth and product innovation. Unlike their measure of investment in AI-skilled workers, our measure of firms' uptake of AI is based on patenting activity, which offers additional evidence on the effects of investing in AI.¹⁰ Our work also provides complementary evidence to the finding in Babina et al. (2024) that AI enables new business ideas and scope expansion. With data on individual workers' skills and transitions, we are able to document that some types of AI significantly increase the hiring of workers with new skills ("scope" augmentation) rather than just increasing the headcount of workers with non-novel skills ("core" augmentation). Thus, we add to the recent literature on corporate value creation via new product markets and innovation (Hoberg and Phillips, 2025; Braguinsky et al., 2021) by demonstrating a plausible channel, i.e., expansion of the scope of worker skills, through which AI can significantly increase firms' growth and productivity.

Finally, our paper adds to a growing literature in financial economics that uses natural language processing (NLP) techniques to study large-scale textual data such as news articles,

⁹ A key assumption underlying our analysis is that many firms do indeed have strong incentives to adopt and use AI technologies upon obtaining AI patent grants. While we believe this assumption to be highly plausible given that a patent grant confers the right to prevent other parties from commercially exploiting an invention, in practice some firms might pursue AI innovation only to profitably license it to others. Nevertheless, a number of facts suggest that AI patent licensing is unlikely to be driving our main results. First, the industrial sector with the most AI patenting activity is manufacturing, not high-tech (see Table 3, Panel A). Second, an examination of patent licensing data (ktMINE License Agreement database) shows that AI patent licensing is rare, totaling fewer than 300 licensing agreements between firms in our sample period. And third, our empirical findings on firm-level productivity and cost efficiency—which are not directly affected by patent licensing—suggest that AI patent licensing is not a major driver of firm value changes.

¹⁰ We acknowledge that some AI technologies are not patented before they are developed or adopted. Nevertheless, focusing on AI patenting activity has the advantage of enabling us to apply generative AI to detailed patent disclosures to discern and study important differences among AI technologies.

corporate announcements, regulatory filings, and call transcripts. Prior studies in this area apply text-filtering or word-embedding algorithms that, while straightforward to interpret, are less able to distinguish shades of meaning. More recently, a nascent branch of the literature has emerged that applies generative AI to the rich textual data in corporate disclosures (see, e.g., Lopez-Lira and Tang, 2023; Li et al., 2024; Jha et al., 2024; and Sheng et al., 2024).¹¹ We contribute to this stream of research by providing the first use of generative AI techniques for systematically identifying and classifying AI innovations from patent texts. Applying generative LLMs in a novel way to millions of patent documents, we are able to classify innovations at scale and with less ambiguity than would be possible with conventional NLP methods. Thus, our work offers a new methodological approach to the study of large-sample textual data that can be potentially applied to many other questions in finance and economics.

The remainder of the paper is organized as follows. In Section 2, we outline a typology for categorizing AI into seven types based on functional capabilities. Section 3 describes our data sources and provides descriptive statistics. Section 4 describes our empirical methodology. In Section 5, we report our results. Section 6 concludes.

2. A Functional Typology of AI

A commonly-held view is that AI consists of computing systems that can perform tasks traditionally requiring human intelligence (Simon, 1977; Korteling et al., 2021). The notion that artificial intelligence can or should do what human intelligence is capable of has featured prominently in the work of computer scientists since at least the mid-20th century. For example, in 1950, Alan Turing proposed a test based on an “imitation game” whereby a computer can be said to possess artificial intelligence if it adequately mimics human behavioral responses under certain conditions (Turing, 1950). In recent years, major technical advances related to artificial neural networks, deep learning, and generative AI have enabled computers to equal or exceed human performance in narrow groups of cognitive tasks (Mijwel, 2021; Griffiths, 2020; Sternberg, 2024). Many computer scientists believe that progress may eventually lead to Artificial General

¹¹ Eisfeldt and Schubert (2024) and Dell (2024) provide recent surveys on the use of generative AI tools in finance and economics research.

Intelligence (AGI), a form of AI that, even without extensive pre-training, would be able to think, reason, create, and generalize as humans do (Goertzel, 2014).

Motivated by the oft-discussed parallels between AI and human intelligence, we propose to classify AI according to key cognitive functions. In particular, we develop a typology that associates each instance of AI with one or more kinds of human-like cognition. Our typology is parsimonious yet comprehensive, enabling us to study the extremely broad array of different possible AI innovations and technologies. Furthermore, because our typology directly accounts for cognitive capabilities that human workers would likely use in performing real-world jobs, it is arguably better-suited for studying labor-market effects compared to other classification approaches that focus on technological components, industrial applications, or specific end uses.¹²

We start by identifying a set of core mental processes that underlie virtually all human cognition. For this purpose, we draw upon the unified theory of cognition proposed by Allen Newell, a prominent computer scientist who is widely considered to have been one of the founders of modern AI.¹³ According to Newell’s theory (Newell, 1990), all human cognitive behavior arises from a set of cognitive mechanisms that exist in six major areas: (1) problem-solving, decision-making, routine action; (2) memory, learning, skill; (3) perception, motor behavior; (4) language; (5) motivation, emotion; and (6) imagining, dreaming, daydreaming. To extend Newell’s theory into a parsimonious framework that can be useful for classifying AI, we gather descriptions of the theorized cognitive mechanisms from the psychology, cognitive science, and computer science literatures. Then, based on a synthesis of the resulting descriptions, we propose seven functional capabilities that can be used to describe and classify any AI system: (1) *Language*; (2) *Perception*; (3) *Motor control*; (4) *Engagement*; (5) *Decision-making*; (6) *Learning*; and (7) *Creativity*.

Three of the cognitive areas in Newell’s theory map in a straightforward way to single functional AI capabilities within our framework. This includes language, decision-making, and

¹² Several recent studies in the literature identify AI patents from keyword searches, machine learning, or USPTO patent classification numbers (see, e.g., Cockburn et al., 2019; Webb, 2019; Alderucci et al., 2020; Yang, 2022; Pairolero et al., 2025), but they do not empirically distinguish or categorize different types of AI according to function.

¹³ A mathematics researcher at the RAND Corporation in the 1950s, Allen Newell later held faculty positions at Carnegie Mellon University in the School of Computer Science, the School of Business, and the Department of Psychology. He played a key role in developing the Information Processing Language (IPL) (1956), a foundational tool for artificial intelligence programming. He also collaborated with Herbert Simon and others to create pioneering AI programs such as the Logic Theorist (1956), which automated mathematical reasoning, and the General Problem Solver (1957), a model of heuristic problem-solving. In 1975, Newell and Simon were jointly awarded the ACM A.M. Turing Award (often regarded as the "Nobel Prize of Computing") for their groundbreaking contributions to the fields of artificial intelligence and human cognition.

learning. Indeed, language not only plays an indispensable role in human cognition, but it also constitutes an important type of cognitive technology in machine intelligence (Carruthers, 2002; Kravchenko, 2007). Decision-making is often described in the literature as another key source of cognition by which an individual can set plans and select rational choices based on a reasoning system (Fox, Cooper, and Glasspool, 2013). And learning is a well-developed process utilized in AI fields (e.g., machine learning, deep learning) that enables individuals to update skills and experience and apply them in different situations over time (Greeno, Collins, and Resnick, 1996; Silver et al., 2021).

In some other cases, we find it useful to map an area in Newell's theory to more than one functional AI capability. For example, we posit that perception and motor control are two distinct but related AI functions. Specifically, the literature describes perception as the reception of stimuli, the extraction and storage of perceptual symbols, and the updating of perceptual systems (Barsalou, 1999). Motor control, in contrast, can be described as the motor center's ability to control, execute, and adjust goal-directed actions based on visual, proprioceptive, and kinesthetic feedback (from a perceptual system) to achieve certain tasks or goals (Grush, 2004; Rosenbaum, 2009).

In yet other cases, a cognitive area in Newell's theory cannot be easily observed or measured as a functional capability of an AI system. In such cases, we specify a single observable AI function that arguably best represents the processes that lie within the cognitive area of Newell's theory. For instance, we posit that engagement (with human users or the environment) is a well-defined AI capability that embodies the "motivation, emotion" mechanisms in Newell's theory. Indeed, the literature argues that motivation sets the stage for engagement, and intrinsic motivation is defined as engagement in a task for the inherent emotions, such as pleasure and satisfaction, that are derived from the task itself (Ryan and Deci, 2000). Similarly, we pinpoint creativity (e.g., the generation or expression of novel content) as an observable AI function that embodies the "imagining, dreaming, and daydreaming" processes laid out in Newell's theory. The literature describes creativity as an important cognitive function that requires novelty enabled by imagination (Pelaprat and Cole, 2011; Stokes, 2014). In other words, imagination facilitates creative discovery by playing a core role in creative tasks.

Based on a paraphrasing of descriptions and definitions from the relevant literature, we can characterize the seven functional AI capabilities in our typology as follows:

- ***Language***: Understanding and using language through speaking, listening, reading, or writing
- ***Perception***: Gathering, organizing, and identifying sensory information from external stimuli to gain an awareness of the environment
- ***Motor Control***: Guiding or controlling the movement of physical objects in response to the environment
- ***Engagement***: Understanding, interacting with, and responding to the behavior of users
- ***Decision-Making***: Evaluating options based on relevant information and subsequently choosing an alternative or a course of action
- ***Learning***: Acquiring new knowledge or conclusions based on evidence, logical reasoning, heuristic reasoning, skills, or values
- ***Creativity***: Transcending traditional ideas, rules, patterns, or relationships to generate new ideas, forms, methods, interpretations, or possibilities

Table 1 summarizes the building blocks of our AI typology, including the seven different functional AI capabilities in Column (1); relevant descriptors from the psychology, cognitive science, and computer science literatures in Column (2); and notable examples of AI applications in Column (3). We acknowledge that real-world applications of AI need not be associated with just one functional capability. For instance, an interactive AI chatbot might be able to process perceptual signals, generate creative content, use natural language to engage with users, and dynamically learn about user preferences. Likewise, an AI-guided autonomous ground vehicle might simultaneously exhibit perception, motor, and learning capabilities. We also note that each type in our typology is meant to capture the degree to which a technological innovation is intended to function as part of an AI system. Thus, for instance, an improved design for a generic microprocessor would not necessarily be considered an AI innovation—even though one possible use of the microprocessor is as a hardware component of an AI system.

3. Data

3.1 Patenting by U.S. Public Firms

We obtain data on firms' innovation from the full-text database of patent applications and grants available at the USPTO Bulk Data Storage System (BDSS).¹⁴ From PatentsView (www.patentsview.org), we further obtain information on individual patents' technology classes as indicated by their International Patent Classification (IPC) codes. Next, we proceed in several steps to merge the USPTO data with data from CRSP/Compustat. The first step is to build links within the USPTO data between patent applications and patent grants. From January 2007 to December 2023, there are a total of 5,913,659 utility patent applications, and 3,765,176 of them are observed to have been granted. Next, from the dataset¹⁵ of Kogan et al. (2017), henceforth referred to as KPSS, we obtain the linking data between USPTO patent grant numbers and CRSP PERMNO (PERMCO) identifiers. (The KPSS data cover 3,160,453 granted patents from 1926 to 2022.) To gather information on the patent application that preceded each grant, we retrieve the internal link between grants and applications from USPTO and match it with the KPSS data. Among patents for which both application information and grant information are available, we have from the KPSS data 1,014,772 patents filed by U.S. CRSP firms.

The previous step does not include (1) patent applications that were not granted by 2022; (2) patent applications that were not granted in 2023; and (3) patent grants that occurred in 2023. To account for such cases, we rely on the mapping between patent assignees and CRSP historical firm names from the previous step to extend the linking between CRSP firms and patents filed in later years. Specifically, from the KPSS data, we use PERMNO (PERMCO) identifiers to retrieve all relevant firms' historical names from the CRSP database. Next, we use patent-level identifying information to retrieve assignee names from (1) the PatentsView database for patent grants and (2) the USPTO bulk data for patent applications¹⁶. Then, for patents in 2022-2023, we check whether their assignee names appear in our matched name pairs to identify additional patents filed by CRSP firms.¹⁷ Using this approach, we are able to identify a further 961,655 patent applications that

¹⁴ <https://bulkdata.uspto.gov/>, accessed in January 2024.

¹⁵ This dataset provides updated data series for KPSS (Kogan, et al., 2017), including patent values and forward citations, a match between patent numbers and PERMNOs, and a match between patent numbers and CPC technology class codes.

¹⁶ For patent applications, we clean and parse the USPTO patent application XML files and extract patent assignee names if they are recognized as organizations.

¹⁷ Specifically, an additional patent is identified when its assignee name can be exactly matched to any assignee name in the pre-existing mapping between CRSP and USPTO.

occurred by 2022, as well as an additional 45,123 patent applications and 21,132 grants occurring in 2023. Finally, we restrict the sample to applications (with available examiner information) made by firms in the CRSP/Compustat merged database. The resulting sample covers 2,307 unique U.S. firms in CRSP/Compustat that filed 1,562,529 patent applications from 2007-2023, of which 1,253,239 were granted by the end of 2023.

3.2 AI-Related Patents

3.2.1 Identifying and Classifying AI Patents Using Generative LLMs

Until recently, the standard approach in finance and economics for analyzing textual data has been to use text filtering or word embedding techniques to classify documents according to single-word frequencies or the co-occurrence of certain pairs of words.¹⁸ While such methods are straightforward and easy to interpret, they do not make full use of the semantic connections between words, sentences, and documents that can dramatically improve textual analysis. The recent development of generative AI tools such as ChatGPT, Claude, and Google Gemini has made it possible to retrieve, categorize, and analyze documents using simple, natural-language queries that refer to general topics or concepts. These instances of large language models (LLMs) are trained on trillions of tokens and can respond to queries using a large set of commonsense knowledge.¹⁹

In view of the difficulties with using traditional NLP techniques to capture the rich semantic content in patent filings, we turn to generative AI to help identify and classify different types of AI innovations. Specifically, we use an advanced, open-source LLM called *Qwen2.5-14b-Instruct*. This generative AI model was developed by Alibaba Cloud and released to the public in September 2024 under an open-source license for broad commercial and research use. The model features 14.8 billion parameters and is an instruction-tuned version of a base model that was trained on data consisting of approximately 18 trillion tokens (<https://github.com/QwenLM/Qwen2.5>). According to several performance benchmarks, as of October 2024, *Qwen2.5-14b-Instruct* was the top-performing LLM in its size class, outperforming many larger, proprietary models. For instance,

¹⁸ Some notable examples of this approach to computational linguistics include Tetlock et al. (2008), Loughran and McDonald (2011), Jegadeesh and Wu (2013), Hoberg and Phillips (2016), and Li et al. (2021).

¹⁹ An emerging literature in finance and economics applies LLMs and generative AI techniques for text classification and content analysis. Eisfeldt and Schubert (2024) provide a review of recent finance research that uses generative AI techniques to analyze and classify textual data.

within the topical area of Engineering, *Qwen2.5-14b-Instruct* was seen to outperform well-known models such as Gemini-1.5-Pro, Claude 3-Opus, and GPT-4 Turbo.²⁰

Our approach to identifying and classifying AI innovations from among the millions of U.S. patent filings during 2007-2023 consists of two main steps. First, for each patent application, we formulate a query to elicit from the LLM an inference about whether the invention is AI-related. To reduce ambiguity and to obtain from the LLM as clear a response as possible, we use prompts that feature a specific yes/no question combined with the text of a patent abstract. For example, to check whether or not U.S. Patent Application #20210209449 is related to artificial intelligence, we prompt the LLM with the following question/instruction, along with the patent abstract as context: *“Is the following invention directly related to artificial intelligence? Respond with just YES or NO. Abstract: Graphical elements in a user interface (UI) may be detected in robotic process automation (RPA) using convolutional neural networks (CNNs). Such processes may be particularly well-suited for detecting ... [remaining abstract text follows]”*. Note that, in this step, we do not require the LLM to provide an explanation for its yes/no inference as doing so would greatly extend the time required for each inference.²¹ Applying this querying procedure to all 5,323,321 patent applications in the sample yields a classification of each patent application as being either AI-related or not.

Next, for each patent application that is deemed to be AI-related, we use seven separate queries to the LLM to ascertain whether or not the application belongs to one or more of the seven AI types. We again use simple yes/no prompts to ensure speed of inference. For instance, to check whether a patent application is related to perception-based AI, we query the LLM with the following prompt: *“Is the following invention directly related to artificial intelligence that has perceptual ability? Respond with just YES or NO. [patent abstract text follows]”*. The overall collection of binary yes/no LLM responses provides a full classification of AI patent applications in our sample.

²⁰ See, e.g., the LLM performance benchmarking leaderboard at <https://huggingface.co/spaces/TIGER-Lab/MMLU-Pro> (accessed October 18, 2024).

²¹ As has been well-documented, token generation accounts for a large majority of the processing time required for an LLM to respond to a query. Our approach of eliciting simple yes/no responses from an LLM without requiring explanations has the advantage of dramatically shortening the overall time required for inference. Moreover, we have verified that directing the LLM to forego providing explanations does not seem to compromise the quality of its inferences.

3.2.2 Validation

We assess the quality and reliability of our LLM-based classifications by employing an “LLM-as-a-judge” approach (Zheng, 2023; Gu, 2025; Li, 2025). In this approach, a subset of the inferences from the *Qwen2.5-14b-Instruct* model are evaluated for correctness by larger, state-of-the-art LLM models. To help ensure the reliability of the assessment, we use as judges some of the strongest available LLMs, including proprietary models from leading developers of generative AI: Google, OpenAI, Anthropic, and DeepSeek. Appendix A provides details on the four “frontier” models that we use as judges.

The first step in the LLM-as-a-judge evaluation is to use *Qwen2.5-14b-Instruct* to conduct in-depth queries for random subsets of patent applications. These queries are similar to the simple yes/no queries used for the overall sample, except that they additionally request an explanation for each inference. For example, to check whether a patent application is perception-based, we prompt *Qwen2.5-14b-Instruct* with the following: “*Is the following invention directly related to artificial intelligence that has perceptual ability? Respond with YES or NO and give an explanation. [followed by the patent abstract text]*”. We then provide each prompt, along with the associated response from *Qwen2.5-14b-Instruct*, to each judge LLM and ask the judge to determine whether the response provided to it is correct or incorrect.

As seen in Appendix Table A.1, an average of about 91.44% of our sampled AI type classifications by *Qwen2.5-14b-Instruct* are correct according to the frontier models. For our AI versus non-AI inferences, 97.23% of sampled classifications are deemed correct by the frontier models. Overall, these results help to confirm the reliability of our inferences regarding which patent texts are related to AI (or to a particular type of AI) and which are not.

To further validate the reliability of our LLM-based approach, we manually review numerous instances of yes/no classifications and explanations from *Qwen2.5-14b-Instruct*. Appendix B provides some illustrative examples for each AI type. Our review of various examples suggests that *Qwen2.5-14b-Instruct* is indeed capable of differentiating between fine shades of meaning in patent abstracts. For instance, consider the case of Patent Application #20230294736, “Offline Tracking System for Autonomous Vehicle Control Systems,” (GM Cruise Holdings LLC), listed in Appendix B. Despite the fact that the abstract for this patent filing does not contain the term “perception” or any other synonymous terms, the model correctly recognizes that the invention demonstrates perceptual AI ability because the disclosed system can track, identify, and annotate

objects in an autonomous vehicle’s environment. Also, we note that the LLM is able to reliably determine when an AI invention does not relate to a particular category. For example, the LLM correctly infers that Patent Application #20210258756, filed by Accenture Global Solutions Ltd., does not directly pertain to engagement with users (see Appendix B). As explained by the LLM, although this invention involves sophisticated data processing and decision-making, its capabilities relate to generating automated alerts for users rather than learning from interactions and engaging with users through AI-driven conversational techniques.

3.2.3 Firm-Level Measures of AI Patenting

For each year in the sample and each of the seven AI types defined in Section 2, we count the number of within-type patents obtained by a firm as well as the number of all other (non-type) patents obtained by the firm.²² The log-transformed versions of these count variables are used to construct the main explanatory variables in most of our regressions. In later tests, we also group the seven different AI patent types into broad categories according to their estimated employment effects. Thus, for example, we construct *Log augmenting (displacing) AI patents* as the log-transformed total number of a firm’s AI patents that belong to types estimated to positively (negatively) impact employment growth in exposed occupations (Table 4, Panel B). Likewise, we construct *Log scope augmenting AI patents* as the log-transformed total number of a firm’s patents that are of types estimated to have a positive effect on the hiring of new skills in exposed occupations (Table 5, Panel B). Section 5.2.1 provides additional details on how we aggregate AI patent types into broader categories to examine labor-augmenting and labor-displacing effects.

3.2.4 Descriptive Statistics: Aggregate and Firm-Level AI Patenting

Figure 1 shows the time-series occurrence of AI patenting by CRSP/Compustat firms from 2007 to 2023. Patent grants are counted as of the disclosure date of the application. In Panel A, the solid curve corresponds to the cumulative monthly frequency of granted AI patents (left scale),

²² Note that we do not count a patent as occurring in the year of grant; instead, we count it in the year during which the underlying patent application was first published or first received an office action from the USPTO. The rationale for focusing on these early dates is that they represent key points in time where substantial new information is revealed about a patent application (i.e., the identity of the patent examiner as well as the nature and potential value of the invention). Indeed, we expect that innovating firms and outside investors may begin to react shortly after a First Office Action date/application publication date by altering their corporate policies and investing behavior. (See the discussion in Section 4 for more details.)

while the dashed curve corresponds to the cumulative monthly frequency of AI patents as a percentage of the cumulative monthly frequency of all patents (right scale). Overall, the graphs show a clear upward trend in AI patenting up until 2020, according to both count and percentage measures. The growth rate of AI innovation starts to diminish in the year 2021, possibly due to firms' financial stress during the COVID-19 pandemic. Over the entire time period from 2007 through 2023, AI patenting accounted for nearly 10% of all patenting by U.S. public firms in our sample. Panel B of Figure 1 shows the time series of cumulative monthly patent frequencies for each of the seven functional AI types that constitute the typology outlined in Section 2. According to the graphs, AI innovations related to decision-making and learning account for the largest fraction of overall AI innovation. These types of AI innovation exhibit higher growth rates throughout most of the sample period compared to the other types. Despite the diminishing growth due to the pandemic in late years, AI innovations related to decision-making and learning appear to undergo a growth in late 2022, likely reflecting the recent mainstream adoption of generative AI tools.

In Figure 2, we report the distribution of firms' AI innovation activity in each type by firm size, labor intensity, and R&D intensity. Panel A reports, by firm size, the percentages of a firm's total patenting that consist of AI patents and of each seven AI patent types in our typology. Consistent with the prior literature (Alekseeva et al., 2021; Acemoglu et al., 2022b; Babina et al., 2024), large firms generate proportionately more AI innovation (8% of total patenting for large firms versus 6% for small firms). Panel B summarizes AI patent shares in labor-intensive and capital-intensive (non-labor-intensive) firms. We find that capital-intensive firms file relatively more AI patents than labor-intensive firms, suggesting that the reduction in labor costs is less likely to be the key motivation behind firms' AI innovation efforts. Panel C compares AI innovation activity in R&D-intensive and non-R&D-intensive firms. As seen in the chart, firms with a higher intensity of R&D spending tend to devote proportionately more of their innovation activity to AI patenting of all types. Nonetheless, the differences between the two groups are not large, consistent with the idea that AI innovation is not necessarily an R&D-intensive process.

We report in Table 2 the distribution of AI patent grants across industrial sectors and IPC patent technology classifications. The industrial sectors that account for the largest numbers of AI patents include Manufacturing (47,440), Transport; storage and communication (46,263), and Electricity; gas and water supply (8,653). Given that most high-tech firms reside within the

communication and electricity sectors rather than in the manufacturing sector, these numbers suggest that AI innovation is a very broad phenomenon that extends beyond the high-tech sector. Panel B reports the distribution of AI patents across the ten most frequently represented IPC technology classes. Most AI patents belong to Computing, Calculating, or Counting (G06), but Electric Communication Technique (H04) and Musical Instruments and Acoustics (G10)²³ also account for substantial fractions of AI patenting.

Table 3, Panel B reports summary statistics for firm-level AI innovation and other financial characteristics. Firms' yearly average number of AI patents is 12.10, which is mainly attributable to decision-making AI (7.23), learning AI (4.98), and engagement AI (4.05). The panel also reports statistics for the four broader categories of AI patents: "augmenting," "scope augmenting," "core augmenting," and "displacing." On average, the frequencies of firms' labor-augmenting and labor-displacing AI patents are 8.78 and 4.09, respectively. The average number of AI patent grants that lead to scope (core) augmentation is 6.45 (7.82) for a firm in a given year.

3.3 Employment Data

3.3.1 Micro-level Data from Revelio Labs

We obtain microdata on a large sample of employees at U.S. firms from a novel database compiled by Revelio Labs. This database provides information on global workforce dynamics and worker characteristics based on public employer and employee records covering more than 4.5 million companies and 1.1 billion resume profiles. With monthly coverage beginning in 2008, the Revelio Labs database details individuals' demographics (e.g., name, gender, ethnicity), job positions (affiliated firms, job titles, job roles, start and end dates, salary levels, SOC occupation codes), skills, education, and job transitions. At the employer level, the Revelio Labs database covers approximately 73% of U.S. public firms recorded in Compustat.

Our analysis of firms' workforce dynamics relies on three key aspects of the Revelio Labs employee-level data. First, being able to observe each employee's occupation and employment history enables us to construct measures of occupation-by-firm-by-year employment changes. In particular, from the individual position file, we identify each employee-employer relationship in a

²³ The Musical Instrument and Acoustics (G10) technology class is not only restricted to musical instruments. It includes technologies applied on sound emitting devices. For example, some artificial intelligence patents on methods or devices for transmitting, conducting, or directing audio can be classified under this category. See more details in https://www.wipo.int/ipc/itos4ipc/ITSupport_and_download_area/20240101/pdf/scheme/full_ipc/en/g10.pdf

given year if the year falls between an individual’s starting and ending year for a job. Using firms’ identity information (i.e., GVKEYs) from the company reference file, we merge the employment data with Compustat and restrict the sample to U.S. public firms. Then, aggregating by SOC occupation codes, we are able to measure yearly employment growth rates for each occupation-firm pair.

Second, we use SOC occupation codes to merge the employment data from Revelio Labs with the Occupational Information Network (O*NET) database (www.onetonline.org). This makes it possible to control for how *a priori* exposed an occupation is to a particular type of AI innovation when studying yearly employment changes at the occupation-by-firm level. As a result, we can test for direct substitutive or complementary effects of a firm’s AI innovation on its employees who work in occupations exposed to the innovation.

Third, we use information from the Revelio Labs data on individual workers’ skill sets. Revelio Labs collects raw skill sets of individuals from online sources (i.e., LinkedIn and other profile resources) and uses a clustering algorithm to build a skills taxonomy that includes hierarchical levels of granularity, ranging from 25 general skill titles to 3,000 specific skill titles. The skills taxonomy is adjusted periodically to adapt to the changing occupational landscape. In our analysis, we use the highest available level of granularity in the taxonomy (3,000 skill categories) and merge the resulting data with job transitions data to identify when a newly hired (departed) worker possesses skills that are novel relative to skills held by the new (former) employer’s incumbent workforce. This enables us to test whether or not firms capitalize on their AI innovation by increasing the hiring of workers with novel skills. Hence, we can distinguish between *scope augmentation*, whereby AI increases demand for human workers with novel skills, and *core augmentation*, whereby AI mainly complements human workers who hold more traditional skills.

3.3.2 Measuring Occupational Exposure to Different Types of AI Innovation

To construct measures of how exposed a given occupation is to the seven different types of AI innovation, we make use of data from the O*NET database on skill requirements underlying each occupation.²⁴ These skill requirements cover 35 different skills, including examples such as

²⁴ Our approach differs from the task-based approaches used by other researchers to study the exposure of occupations to new technologies (e.g., Acemoglu and Autor, 2011; Brynjolfsson et al., 2018; Eifeldt et al., 2025).

“Critical Thinking,” “Social Perceptiveness,” “Reading Comprehension,” “Judgment and Decision-Making”, and “Programming.” We manually review each of the 35 skill requirements to determine if the skill is fundamentally related to one or more of the seven cognitive AI capabilities in our typology. Some of the skills link in an obvious way to one or more of the capabilities. For instance, the O*NET skill “Judgment and Decision-making” maps immediately to the capability of *decision-making*, while the O*NET skill of “Writing” clearly links to the capability of *language*. However, in some cases, it is not immediately obvious whether a skill maps to a particular cognitive capability. In such cases, we consider a link to be present if the cognitive capability is an essential requirement for a human worker to be able to apply the skill in the workplace.

We combine our mappings from skill requirements to AI capabilities with O*NET²⁵ information on the importance of the skill requirements to gain an overall picture of whether each 3-digit occupation is exposed to each type of AI innovation. For example, if a 3-digit occupation has at least 4 out of 35 (more than 10%) required skills that map to perceptual cognition, and if each of the skills has high importance (O*NET importance score ≥ 3), then we consider the occupation to be exposed to perception-based AI. Table C.1 in Appendix C shows, for each AI type, the top three most-exposed 3-digit SOC occupations. An occupation’s degree of exposure to an AI type is calculated as the occupation’s weighted count of skill requirements in the O*NET database that map to the AI type, where the weights are the O*NET importance scores of skill requirements. The table shows considerable heterogeneity in which occupations are most exposed to a given AI type. Nonetheless, some systematic patterns are evident. For example, Top Executives (SOC Code 111) is the occupation that is most exposed to both Creativity AI and Language AI. Occupations that involve supervision are present among the top three most-exposed occupations with respect to all seven AI types.

3.3.3 Occupation-Level Changes in Employment and Worker Skills

Such studies quantify an occupation’s exposure to a new technology by using job task descriptions, i.e., descriptions of specific worker actions performed on the job. One limitation to using task descriptions to quantify AI exposure is that job tasks for the same occupation may change significantly across time periods, which is often not reflected in the O*NET data. By comparison, skill requirements for an occupation may remain relatively constant over time, providing a more stable means of linking occupational requirements with functional AI capabilities.

²⁵ In the O*NET database, each occupation at the most disaggregated (8-digit) SOC level is associated with numerical importance weightings that indicate how essential each of the 35 required skills is to the occupation. These scores range from 1 to 5, with higher scores indicating greater importance. To simplify the aggregation of skill requirements at the occupation level, we code any skill having a score of 3.0 or above as “important” to the occupation, while any skill with importance below 3.0 is deemed “not important.”

We construct our main employment outcome variables at the occupation-by-firm-by-year level (where occupations are at the 3-digit SOC level of disaggregation). Specifically, using the Revelio Labs individual position file, we calculate *Employment Growth* as the yearly headcount change in a 3-digit SOC occupation within a firm, divided by the firm’s scaled total headcount in the prior year.

To analyze labor augmentation effects that expand the scope of skills within a firm’s workforce, we define *Employment Growth (Novel Skills)* as the yearly change in employment of workers with novel skills, measured at the occupation-by-firm level and divided by the firm’s scaled total headcount in the prior year. We ascertain which workers have novel skills by merging the Revelio Labs individual position file with individual skills data. (See Table C.2 in Appendix for examples of individual skill sets for both newly-hired and newly-departed employees.) Specifically, for each worker with available yearly occupation and firm information, we compile his or her skill set vector from a total of 3,000 possible skills in the data. Comparing each worker’s skill set vector against the element-wise median skill vector among all workers in the same occupation and firm, we deem the worker to have novel skills if they have one or more skills that are absent from the median occupational worker’s skill set.

Panel A of Table 3 shows summary statistics for occupational employment growth and the share of the firm’s total employment within occupations exposed to each type of AI. The mean (median) occupational employment growth rate is positive (zero), indicating that the typical occupation within firms does not experience a net outflow of employees over time. However, the mean growth rate in occupation-level hiring of employees with new skills is negative, suggesting an overall bias towards increasing scale rather than scope. Motor AI, Creativity AI, and Learning AI each has a broad scope of potential impact on employment: on average, more than 70% of employees work in occupations that are exposed to these AI types.

To gain an initial understanding of how AI innovation might affect employment, we explore how firm-level employment within different occupations changes in response to each type of AI innovation. Figure 3 shows a heatmap that visualizes the general directions of employment changes—i.e., positive or negative—associated with different combinations of AI type and occupational group. To construct the heatmap, we first identify cases in which a firm’s AI patenting in year t is dominated by a single type that constitutes more than one-third of all the firm’s AI patents during the year. We sort occupation-firm-year observations into groups according to year,

firm-size quartile (total assets), 3-digit SOC occupation, and predominant AI type. Then, for each observation, we calculate an employment growth score based on the within-group quartile ranking of the observation's year $t+1$ employment change (1 = bottom quartile, 4 = top quartile). To construct the cells of the heatmap, we standardize employment growth scores to lie between -1 and 1 and average them within pairs formed by AI type and 2-digit SOC occupational group. Darker red in a cell indicates that the relevant AI type leads to more positive employment changes in the relevant occupational group, while darker blue indicates more negative employment changes.

Figure 3 reveals a few striking patterns in the data. First, different occupation groups appear to respond very differently to AI innovation. For example, all seven AI types lead to positive employment growth for some occupation groups, such as Architecture and Engineering; Life, Physical, and Social Science(s); Arts, Design, Entertainment, Sports, and Media; Business and Financial Operations; Computer and Mathematical; and Management. However, all seven AI types lead to negative employment growth for three other occupation groups: Building and Grounds Cleaning and Maintenance; Farming, Fishing, and Forestry; and Healthcare Support. Second, the heatmap suggests that generative AI technologies, which have surged in popularity and entered the mainstream during the past few years, may in fact lead to mixed employment effects. On the one hand, generative AI relates to learning, decision-making, and creativity, which are among the favorable AI types for subsequent employment growth according to the heatmap. On the other hand, since it embodies language AI capabilities, generative AI may be associated with negative employment growth in “pink-collar” and “white-collar” occupations (e.g., Community and Social Service occupations; Legal occupations) as well as in “blue-collar” occupations involving manual labor (e.g., Installation, Maintenance and Repair; Building and Grounds Cleaning and Maintenance occupations). Third, of all the AI types, motor control is the one that is most strongly associated with negative occupational employment growth. Interestingly, while motor AI appears to lead to negative employment growth in Legal occupations and nearly all blue-collar occupations, it is associated with positive employment growth in many white-collar professions.

4. Empirical Design

To study the effects of AI innovation on employment growth, we use detailed occupational employment data and account for the fact that some occupations are a priori “exposed” to some types of AI innovation, while others are not. Specifically, we estimate a series of panel regressions

using a dataset that combines information on firm-level patenting with information on employment changes at the firm-by-occupation level. Each regression links a given type of a firm’s AI patenting to the firm’s subsequent employment growth in exposed occupations—while also controlling for the possible influence of the firm’s other patenting activity. Thus, our empirical design aims to isolate and distinguish the direct effects (either augmenting or displacing) of a particular AI innovation type from any indirect effects that might be caused by either non-AI innovation or non-focal types of AI innovation.

For each type k of AI among the seven in our typology, we estimate a baseline regression of the following form:

$$y_{i,j,t+s} = \beta_k Exposed_{j,k} \times TypePatents_{i,k,t} + \gamma_k Exposed_{j,k} \times NontypePatents_{i,k,t} + \delta_k (1 - Exposed_{j,k}) \times AllPatents_{i,t} + \lambda'_k X_{i,t-1} + \theta_i + \rho_{j,t} + \mu_{i,t} + \alpha_{i,t} + \epsilon_{i,j,k,t} \quad (1)$$

where $y_{i,j,t+s}$ is the change in occupation j ’s employment at firm i between years t and $t + s$; $Exposed_{j,k}$ is a binary variable equal to one if occupation j is exposed to AI type k according to our skills-based mapping (Section 3.3.2); $TypePatents_{i,k,t}$ is the log of one plus the count of granted type- k AI patents applied for by firm i in year t ; $NontypePatents_{i,k,t}$ is the log of one plus the number of all other granted patents applied for by firm i in year t ; and $AllPatents_{i,t}$ is the log of one plus the count of all granted patents applied for by firm i in year t .²⁶

The vector $X_{i,t-1}$ denotes a set of lagged control variables including firm size, firm performance, market-to-book ratio, and R&D expenditures. θ_i , $\rho_{j,t}$, and $\mu_{i,t}$ represent firm fixed effects, occupation-by-year fixed effects, and industry-by-year fixed effects, respectively. Apart from absorbing any unobserved, time-varying factors that are common to all firms and occupations, these fixed effects control for time-varying shocks that are specific to individual occupations or industries. The term $\alpha_{i,t}$ represents fixed effects for the year-specific counts of

²⁶ Following Farre-Mensa et al. (2020), we count a patent application as being in year t if its USPTO “First Office Action” date occurs during the year. The First Office Action date for a patent application is the date on which the USPTO contacts the applicant by mail about the examiner’s initial examination and about material deficiencies, if any, that must be remedied for allowance to a grant to occur. This date corresponds to the earliest date on which the applicant learns the identity of the examiner (and hence becomes informed about the likelihood that the patent will be granted).

patent applications filed by firms; it accounts for the obvious fact that granted patents are correlated with the volume of a firm’s applications (which can itself be correlated with other firm characteristics). We cluster standard errors at the 2-digit SIC industry level to account for possible within-industry correlations across occupations and firms. The main coefficient of interest, β_k , captures the extent to which more AI innovation of type k leads to more augmenting (positive) or more displacing (negative) employment growth.

A key challenge in using the above specification to identify the effects of AI innovation is that omitted variables, not innovation activity per se, might be the true driver of employment changes. For example, competitive forces might cause firms to invest heavily in AI technologies and, at the same time, to embark on cost-cutting programs that lead to lower future employment growth. In this scenario, the regressions in (1) would yield biased and inconsistent estimates of the impact of greater AI innovation on employment growth. To address this potential endogeneity problem, we make use of plausibly exogenous variation in the assignment of patent applications to USPTO patent examiners. This identification approach, which builds on the work of Gaule (2018), Sampat and Williams (2019), and Farre-Mensa et al. (2020),²⁷ relies on two key institutional features. First, individual patent examiners differ systematically in their leniency, i.e., in how likely they are to allow an application into granted status. Second, within an art unit, the allocation of patent applications to individual examiners is largely random (Lemley and Sampat, 2012; Sampat and Williams, 2019).²⁸

We extend and generalize the approach of Farre-Mensa et al. (2020) to account for different groups of patenting by the same firm in the same year.²⁹ First, for a patent application p assigned to an examiner j in year t , we estimate examiner leniency, $E_{p,j,t}$, as the fraction of all other applications that the examiner reviews during the year that are (eventually) granted.³⁰ Then, for

²⁷ Gaule (2018) uses variation in USPTO examiner leniency to study the effects of patenting on the success of startup firms. Sampat and Williams (2019) employ a similar approach to explore whether genome patents affect scientific investment and follow-on innovation. In Farre-Mensa et al. (2020), variation in patent examiner leniency is used to study how winning a patent grant affects a startup’s employment and sales growth.

²⁸ From written interviews with USPTO examiners about the patent assignment process, Lemley and Sampat (2012) find no evidence of systematic assignment other than that based on technological attributes that determine an application’s appropriate USPTO art unit.

²⁹ The analysis of Farre-Mensa et al. (2020) only considers a firm’s first patent grant and does not distinguish between different patent types or categories. By accounting for all of a firm’s patent applications in a given year, we are able to distinguish the effects of AI and non-AI patents as well as the effects of different types/categories of AI patents.

³⁰ This “leave-one-out” approach ensures that the leniency for a particular patent application is not contaminated by whether the application itself is granted or rejected by the examiner to which it is assigned.

any given set of patent applications filed in a year by a firm, we can calculate total leniency by simply summing up the leniencies associated with the various applications. In essence, the total leniency for a firm's set of patent applications represents how "lucky" the firm is in drawing examiners who are more likely to give favorable assessments.

With examiner leniency measures in hand, we can use instrumental variables estimation to address the potential endogeneity. Specifically, for each AI type k , we use Two-Stage Least Squares (2SLS) to estimate Equation (1). In the model, there are three endogenous variables: $Exposed_{j,k} \times TypePatents_{i,k,t}$, $Exposed_{j,k} \times NonTypePatents_{i,k,t}$, and $(1 - Exposed_{j,k}) \times AllPatents_{i,t}$. Hence, we require at least three separate instruments for valid identification. To proceed, we first calculate $Z_{i,k,t}^{Type}$, $Z_{i,k,t}^{NonType}$, and $Z_{i,t}^{All}$, which are the year- t total leniencies for firm i 's type- k patents, its non-type- k patents, and all of its patents, respectively. We then interact these total leniencies with exposure measures to form three instruments: $Exposed_{j,k} \times Z_{i,k,t}^{Type}$, $Exposed_{j,k} \times Z_{i,k,t}^{NonType}$, and $(1 - Exposed_{j,k}) \times Z_{i,t}^{All}$. Because we estimate the model via standard 2SLS, there are three first-stage regressions (one for each endogenous variable), and each regression includes the three instruments as right-hand side variables, along with the same controls and fixed effects appearing in the main equation (1). Since the endogenous variables and instruments are equal in number, they constitute a just-identified model, and 2SLS can give consistent estimates without suffering from overidentification issues.

Based on the coefficient estimates for β_k in Equation (1), we infer whether AI innovation of type k has labor-augmenting effects, labor-displacing effects, or neither. We can then group the individual types of AI patents into broad categories, such as labor-displacing AI or labor-augmenting AI, and study the effects of these categories on firm-level outcomes. Specifically, to examine the impact of an AI innovation category on firm-level outcomes, we use 2SLS to estimate the following:

$$y_{i,t+s} = \beta_k CategoryPatents_{i,k,t} + \gamma_k OtherPatents_{i,k,t} + \lambda'_k X_{i,t} + \theta_i + \mu_{i,t} + \alpha_{i,t} + \epsilon_{i,k,t} \quad (2)$$

where $y_{i,t+s}$ is a firm-level outcome of interest (e.g., change in productivity, operating costs, or market valuation), measured as of year $t + s$. The endogenous explanatory variables are $CategoryPatents_{i,k,t}$ (the log of one plus the count of granted category- k patents applied for by firm i in year t) and $OtherPatents_{i,k,t}$ (the log of one plus the number of all other granted patents applied for by firm i in year t). The instruments for these variables are $Z_{i,k,t}^{Category}$ and $Z_{i,k,t}^{Other}$, which are the year- t total examiner leniencies for firm i 's category- k patents and other patents, respectively.

Note that the model in Equation (2) pertains to firm-year observations and thus is considerably simpler than the model in Equation (1). Indeed, the regressions for Equation (2) do not rely on measures of occupation-specific exposure to the various AI patent types, and thus they do not require occupation-by-year fixed effects ($\rho_{j,t}$). We cluster standard errors at the 2-digit SIC level to account for possible within-industry correlations across firms. As was done previously for Equation (1), we generally count a patent application as being in year t if its USPTO First Office Action date occurs during the year.

Further, as discussed in Farre-Mensa et al. (2020), it should be acknowledged that IV-2SLS estimates can differ substantially from those of the OLS model due to differences between local average treatment effects (LATEs) and average treatment effects (ATEs). Specifically, the 2SLS instrumental variables approach only captures the LATE, and its estimates identify the effects of AI patents for the subsample of firms whose patent grants are determined at the margin by examiner leniency. AI patent applications of extremely low and high quality are not expected to be responsive to our instrument as their patent examination outcomes are more obvious. Thus, OLS estimates could be either larger or smaller than 2SLS estimates, depending on the difference between LATE among firms that file marginal patents and ATE among all firms in the sample.

5. Results

In this section, we first report our main findings on the occupation-by-firm employment effects of different types of AI innovation. We then group the seven functional AI types into broad categories based on whether their employment effects are estimated to be positive or negative. Based on this grouping, we study the effects of broad categories of AI on various firm-level outcomes, including value, productivity, and costs.

5.1 Do AI Technologies Complement or Substitute for Human Workers?

5.1.1 *The Labor-Augmenting and Labor-Displacing Effects of AI Innovation*

Our empirical analysis starts with the effects of different AI technologies on occupational employment growth. New AI technologies may increase firms' demand for human workers to adapt to new production processes and to operate advanced systems (Acemoglu and Restrepo, 2019). However, AI innovations may also substitute for human workers, leading to technological unemployment (Frey and Osborne, 2017). Both of these views can be true if AI technologies have different functional capabilities that affect labor growth differently. For instance, an AI-based user interface implemented in the workplace could be used by employees to assist with collecting and delivering information for use in interactions with customers. This type of interface could significantly increase the efficiency of existing workers without displacing their jobs. In addition, the AI technology could also increase firms' demand for other high-tech workers, such as network engineers who handle data security and system maintenance. Nonetheless, there are other AI technologies, powered by machine learning and NLP algorithms, that could displace a firm's workforce by performing analysis and logical reasoning more efficiently and at lower cost than the average data-centric human worker could.

To explore the effects of AI technologies based on their functional capabilities, we implement, for each of the seven AI categories, an IV-2SLS approach as described in Section 4.2. We measure firm-level AI innovations as the log-transformed number of patent grants in a particular AI type each year. Our primary focus is on the direct substitutive and complementary effects of AI on jobs, and hence we rely on measures of AI exposure at the technology-occupation level constructed in Section 3.3.2. By investigating the relation between AI innovation and firm occupational employment changes among exposed occupations, we can isolate the direct substitution and complementarity effects of AI from other, confounding effects that could be contributing to increases or decreases in labor demand through indirect channels.

Table 4 reports baseline OLS regression results (Panel A) and 2nd-stage 2SLS regression results (Panel B) examining the effects of a type of AI innovation on overall employment growth in exposed occupations. Kleibergen-Paap rk Wald F statistics for the 2SLS regressions are significant, rejecting the hypothesis of weak instruments. The dependent variable is *Employment Growth*, calculated as the change in employee headcount in an occupation within a firm divided

by the firm's total number of employees in the prior year. The *Exposed* dummy indicates whether a given occupation requires important skills that map to the relevant type of AI (see Section 3.3.2). We include lagged controls for firm size (total assets), ROA, Market-to-book, and R&D spending. Each regression includes firm fixed effects, (2-digit SIC) industry-by-year fixed effects, (3-digit SOC) occupation-by-year fixed effects, and fixed effects for the interaction of year and a firm's total patent applications during the year.

Panel A shows significant positive coefficients on *Exposed* $\times$ *Log type- k patents* in Columns (4)-(6), suggesting that AI patenting related to engagement, decision-making, and learning is positively associated with employment growth in exposed occupations. In Panel B, the significant positive coefficients on the *Exposed* $\times$ *Log type- k patents* in Columns (1), (4), (5), (6), (7) suggest that AI patents related to language, engagement, decision-making, learning, and creativity complement human workers in exposed occupations. Learning AI patents have larger economic significance than other types of AI patents. Specifically, a 1% increase in firm-level learning AI patent grants leads to an increase in employment growth by 0.02 percentage points. The significant negative coefficients on *Exposed* $\times$ *Log type- k patents* in Columns (2) and (3) show that AI innovations related to perception or motor tend to displace workers in exposed occupations. In particular, a 1% increase in firm-level motor AI patent grants leads to a decrease in employment growth by 0.019 percentage points.

A comparison of the estimates from OLS and 2SLS regressions in Table 4 shows that the estimated OLS coefficients are generally smaller in magnitude. This likely reflects two possible effects. First, OLS may underestimate the impact of AI patenting on employment growth due to omitted variables bias. For example, competitive pressures may simultaneously push firms to file more AI patents and to downsize their workforce. In this case, OLS underestimates the employment effect of AI patents because a portion of the variation in observed layoffs reflects competitive pressure rather than the causal impact of patenting. Second, consistent with the explanation given in Farre-Mensa et al. (2020), the local average treatment effects (LATEs) identified in the 2SLS models are larger than the average treatment effects (ATEs) in OLS.

Our findings so far reveal that different AI innovation types can have very different—even qualitatively opposite—effects on employment. What explains this heterogeneity of effect? We can interpret the heterogeneity in terms of the literature that differentiates between functional capabilities corresponding to high and low cognitive levels (O'Reilly, 2006; König et al., 2013).

On the one hand, high-level cognitive skills (i.e., language, engagement, decision-making, learning, and creativity) can enable complex and abstract tasks. On the other hand, low-level cognitive skills (i.e., perception and motor control) are more environmentally dependent and tend to focus on passive responses to external stimulus.³¹ Therefore, we expect that AI types related to high-level cognition will generally be used to assist or collaborate with human workers³², whereas AI related to low-level cognition can more easily replace human workers in codifiable, automatable job roles. To summarize, our occupation-specific employment analysis confirms that AI innovations can have either substitutive or complementary effects on human workers depending on which type of AI is involved, which occupation(s) are exposed to that type of AI, and whether the AI focuses on high-level or low-level cognition.

5.1.2 Labor Augmentation: Scale Versus Scope

Next, we unpack the labor augmentation effects and examine them on both the intensive and extensive margins. As discussed earlier, one way that AI innovation can augment labor is by complementing and leveraging worker skills within current roles (Acemoglu and Autor, 2011; Acemoglu et al., 2022b), thereby prompting scale expansion through hiring of existing skills in the core business. However, AI innovation could also lead to new products, new businesses, and expansion of scope, which may necessitate the hiring of workers with novel skills (Cockburn et al., 2019; Babina et al., 2024).

To explore whether an AI type augments a firm’s workforce by increasing the scale of labor or by increasing the scope of hired skills, we focus on the five AI types that are seen to increase headcount growth as reported in Table 4. We measure *Employment Growth (Novel Skills)* as the change in the number of employees in an occupation who have new skills, divided by the prior year’s total number of employees within the firm. We define an employee as having new skills if

³¹ Perception and motor AI mainly enable routine physical tasks that are instructed and repetitive (Fod, Matarić, and Jenkins, 2002). These tasks are less complicated and usually can be done by an AI system independently. The real-world applications of these AI systems (e.g., face recognition, autonomous vehicle driving) often relate to automation which disrupts labor markets when human roles are more easily codified and replaced.

³² AI systems used in healthcare diagnostics or legal proceedings, such as systems related to language, learning, and decision-making, require human collaboration to interpret outputs and ensure accountability. For example, algorithmic recourse frameworks integrate Bayesian preference elicitation to generate personalized explanations for AI decisions (e.g., loan denials). These frameworks depend on human feedback to balance technical feasibility with real-world constraints, such as socioeconomic factors. Similarly, human-AI collaborative recommender systems may use hybrid models in which human moderators adjust AI suggestions to reduce biases and enhance user trust.

his or her skill set contains unique skills that are absent from the occupational median skill set. Panels A and B of Table 5 separately report OLS regression results and 2SLS (2nd-stage) results examining the effects of AI innovation on the net hiring of employees with new skills. The dependent variable is *Employment Growth (Novel Skills)*, and all other variables are defined as in Table 3. In Panel A, we find that the coefficients on the *Exposed* $\times$ *Log type- k patents* in Columns (2), (4), and (5) are significantly positive, indicating that AI patents related to engagement, learning, and creativity are associated with increases in the scope of worker skills. We find no significant effects of language AI and decision-making AI on skill accumulation, suggesting that language AI and decision-making AI mainly complement workers with existing skills. Using 2SLS estimation in Panel B to address potential endogeneity, we find results consistent with those from OLS.

The evidence in this section indicates that, even among AI types that complement a firm's workforce with high-level cognitive processes, different types can have very different effects on firms' labor demand. These differences can be interpreted in terms of contrasting cognitive styles of AI systems. In particular, the literature on cognitive architectures draws a distinction between *adaptive* and *innovative* cognitive styles (Kilicay-Ergin and Jabllokow, 2012). Learning, creativity, and engagement constitute more innovative forms of cognition that can generate novel outputs based on real-time feedback, such as what is involved in human supervision (Rodd et al., 1992). In contrast, language and decision-making forms of cognition are more adaptive since they mainly generate predictable output through pattern recognition (Agrawal et al., 2018). This contrast in cognitive styles can help explain why we find that AI innovations related to learning, creativity, and engagement tend to increase demand for new skills, whereas AI innovations related to language and decision-making do not.

Overall, the results in Tables 4 and 5 are consistent with the theme in the recent literature that technology in general has both complementary and substitutive effects on labor demand. Also, the finding that different types of AI innovation can have very heterogeneous effects on occupational employment helps provide a more nuanced perspective in the ongoing debate over whether AI technology ultimately drives labor augmentation or labor displacement.

5.2 AI Innovation and Firm Efficiency

Our previous findings suggest that AI innovation significantly impacts firms' employment dynamics by both complementing and substituting human workers in exposed occupations. At the

firm level, labor augmentation and labor displacement can involve different cost-benefit tradeoffs, leading to implications for firm efficiency. In this section, we explore some important real and financial effects of workforce changes brought about by AI innovation. We expect labor augmentation and labor displacement to affect firm efficiency in different ways. For example, if a firm adopts an AI assistant for potential use by employees, the firm's efficiency can be increased due to improvements in worker productivity. At the same time, in the case of reduced labor demand that leads to substitution of AI technologies for human workers, firms might benefit mainly from lower wage costs.

5.2.1 *Augmenting and Displacing Categories of AI Patents*

Based on the regression results in Panel B of Table 4, we categorize perception AI patents and motor AI patents as *displacing* since these types of AI patents negatively impact employment growth. In contrast, the other five types of AI patents are deemed to be *augmenting* because they result in higher employment growth. We also further specify subcategories of augmenting AI patents based on how such patents affect firms' acquisition of new skills. In particular, we categorize AI patenting related to engagement, learning, or creativity as being *scope augmenting* since each of the three patenting types increases the net hiring of workers who have novel skills (see Panel B of Table 5). The two remaining types, namely, language and decision-making, lead to increased overall headcount (Table 4, Panel B) but do not significantly boost the net hiring of workers with novel skills (Table 5, Panel B). Therefore, we categorize language AI and decision-making AI as *core augmenting*. Figure 4 illustrates the categorization of the seven AI types and the associated in-sample patent count frequencies for each type.

5.2.2 *The Effects of AI Innovation on Firm Productivity*

Using our broad grouping of AI patents into augmenting, scope augmenting, core augmenting, and displacing categories, we study how AI innovation impacts firms' productivity. As discussed in the literature (Solow, 1957; Nadiri, 1970), Total Factor Productivity (TFP) increases when a firm uses its inputs (i.e., capital and labor) effectively. We use the growth rate of total factor productivity (TFP) as a measure of changes in firm efficiency. Building on the model from Solow (1957), we estimate firm-level TFP as the residual from a firm-year level linear production function model with capital and labor inputs. Firm-level production inputs in the

residual estimation regression are proxied for by property, plant, and equipment (PPENT) and total employment (from Revelio Labs).

Table 6 reports the OLS regression results and the 2SLS (2nd stage) results examining the effects of AI innovation on firm productivity. The sample consists of firm-year observations with non-missing patent examiner information. The dependent variable, *TFP growth*, is the percentage change in TFP from year t to $t+1$. Columns (1)-(4) report the results of OLS regressions, and Columns (5)-(8) report second-stage 2SLS results. The main explanatory variables in the regressions are *Log category-k patents* and *Log other patents*. For the 2SLS regressions, we construct instruments for the two explanatory variables based on measures of patent examiner leniency as detailed in Section 4.

From the OLS results in Table 6, displacing AI patents appear to be positively and significantly associated with firms' subsequent productivity growth. In contrast, there appears to be no significant relation between other categories of AI patenting and changes in TFP. However, once we account for endogeneity in the 2SLS regressions, the picture changes: augmenting AI patents, including both scope and core augmenting AI patents, all have significant positive effects on subsequent TFP growth. Nevertheless, there is no causal evidence to indicate that displacing AI innovations help increase firms' productivity.³³ This suggests that if displacing AI innovations (i.e., those related to perception and motor control) do bring significant value to innovating firms, then likely channel for this is via lowered production costs rather than increased top-line performance.

5.2.3 The Effects of Displacing AI Innovation on Cost Efficiency

To further investigate the effects of labor-displacing AI innovation, we examine changes in firms' cost efficiencies. Table 7 reports the results of OLS and 2SLS regressions that relate displacing AI innovation to two distinct cost measures: the change from year t to year $t+1$ in (log-transformed) selling, general, and administrative expenditures (XSGA) and in (log-transformed) operating expenses. The main explanatory variables in the regressions are *Log displacing patents* and *Log non-displacing patents*, which are the log-transformed year- t counts of the firm's displacing and non-displacing patents, respectively. As seen in the table, the OLS regressions reveal either insignificant or significantly positive coefficient estimates for *Log displacing*

³³ The upward bias in OLS estimates for displacing AI might be due to some unobserved omitted variables (e.g., managerial quality) that both boost firm productivity and, at the same time, reduce labor inputs through AI innovation.

patents.³⁴ However, Columns (2) and (4) show that, once endogeneity is accounted for via instrumental variables estimation, negative coefficients for *Log displacing patents* emerge. This finding is consistent with the view that the main benefit firms obtain from pursuing labor-displacing AI innovations takes the form of wage cost reductions.

5.3 AI Innovation and Firm Value

5.3.1 Overall Effects of AI Innovation on Firm Value

Next, we examine the implications of AI innovation for firm value. We expect AI innovation to bring net value increases to firms through either productivity enhancements or, in the case of displacing innovations, cost reductions. Table 8 reports the results of regressions relating the net change in Tobin's Q from year t to $t+1$ to AI innovation. As in previous tests, the main explanatory variables are *Log category- k patents* and *Log other patents*, instrumented by measures of patent examiner leniency as described in Section 4. Columns (1)-(4) report the results from OLS regressions, and the other columns report the second-stage 2SLS regression results. The significant positive coefficients for category- k patenting in Columns (5)-(8) show that augmenting AI and displacing AI alike have significant positive effects on firm value. Moreover, within the broader category of augmenting AI patents, both scope-augmenting and core-augmenting patents yield positive value effects. The economic magnitude of the value creation by AI patents is also sizeable. On average, a 1% increase in labor-augmenting (labor-displacing) AI patents at the firm-level leads to an increase of 0.035 (0.041) in the value change from year t to $t+1$. Taken as a whole, the evidence shows that AI innovations systematically increase innovating firms' value, likely through channels that are particular to the underlying types of innovation being pursued.

5.3.2 Heterogeneous Value Effects of AI Innovation

In this section, we further explore the value effects of AI innovation by conditioning on different measures of labor market frictions that firms may face. Specifically, we study (1) how the value effects of labor-augmenting AI depend on a firm's hiring frictions and (2) how the value

³⁴ For labor-displacing AI, the estimated coefficient is significantly positive in the OLS regression, suggesting potential bias from unobserved omitted variables. For example, a positive demand shock that results in a surge in orders may prompt a firm to invest more in labor-displacing AI to meet the increased demand without expanding the workforce. At the same time, the firm might incur higher operating expenditures (e.g., materials, logistics, marketing) to fulfill the higher volume of orders.

effects of labor-displacing AI are shaped by the extent of potential wage-cost savings available to a firm.

As discussed in the previous section, AI innovation can increase firm value through labor augmentation because new technology improves the firm's growth by complementing employees (who may or may not have novel skills). Given that the costs of external hiring are higher than the costs of internal promotions (Hamermesh and Pfann, 1996), the potential value effects of labor-augmenting AI can depend critically on the state of the external labor market, especially when AI increases firms' demand for new skills. We expect that, in general, augmenting AI innovations are more value-creating when firms have lower costs associated with hiring activities.

To conduct subsample analyses, we rely on three measures of external hiring costs. First, we use the state-level employee turnover rate as a proxy for the ease of hiring in the focal firm's external labor market. Firms located in areas with a high rate of employee turnover may benefit from robust growth in the local labor market supply and a lower cost of hiring new employees (Jovanovic, 1979). Second, we use the industry-level rate of labor unionization as another measure of external hiring frictions. According to the literature, labor unions tend to reduce workers' mobility by increasing their job satisfaction and attachment (Freeman and Medoff, 1984). Third, we use the state-level enforceability of noncompete agreements (NCAs) as another measure of external hiring frictions. Prior studies have documented that an increase in noncompete enforceability can restrict labor mobility (Starr et al., 2021; Jeffers, 2024), thus inducing frictions into firms' hiring processes.

Table 9 reports 2SLS regression estimates for tests involving labor market frictions. The dependent variable is the net change in Tobin's Q from year t to $t+1$, and the main explanatory variables are *Log category- k patents* and *Log other patents* (instrumented by measures of patent examiner leniency), where k captures either augmenting, scope-augmenting, or core-augmenting AI patents. Panel A reports regression results for subsamples of firms headquartered in states with high (above the sample median) employee turnover rates and those headquartered in other states. The employee turnover rate is defined to be the net job transitions (inflow minus outflow) as a percentage of total employment within the state. In Columns (1)-(3), we find significant positive effects of all three augmenting categories on firm value when the in-state turnover rate is high. In contrast, when the in-state turnover rate is low, none of the categories of patenting has a significant effect on firm value. These results point to low hiring frictions in the external labor market as a

critical factor that enables firms with AI innovations to benefit from their labor-augmenting innovations.

Panel B reports the results of subsample regressions for firms in industries with a high (above the sample median) worker unionization rate and those in industries with a low unionization rate. The unionization rate is the industry-level percentage of union members (drawn from <https://www.unionstats.com>). Consistent with the view that union protection increases hiring frictions for firms, all three categories of augmenting AI innovations have significant positive effects on firm value (Columns (4)-(6)) when a firm operates in a low unionization rate industry. Different from core-augmenting AI, scope-augmenting AI has significant positive value effects regardless of high or low unionization rates. This is consistent with the notion that firms with scope-augmenting AI are less vulnerable to industry-level hiring frictions because they tend to focus on novel skills that can be accumulated across industries.

Panel C reports the results of subsample regressions for firms headquartered in states with high (above the sample median) NCA enforceability and for firms headquartered in states with low (below the sample median) NCA enforceability. We obtain the NCA enforceability data from 1991 to 2014 from Johnson et al. (2025). We extend the data through the remainder of our sample period (2023) by imputing the missing data using the most recent NCA enforceability score in each state. In Columns (4) and (6), we find that augmenting and core-augmenting AI innovations have significant positive effects on firm value only when a firm is headquartered in a state with low NCA enforceability. Interestingly, we do not observe a similar pattern for scope-augmenting AI patents: they have significantly positive value effects regardless of whether the within-state NCA enforceability is high or low. A plausible explanation is that scope augmentation increases demand for new skills that can be accumulated from non-rival firms or from other states, and these sources of hiring are less affected by in-state NCA enforceability.

The tests in Table 9 show that firms gain more from augmenting AI when overall labor mobility is high, but they do not account for the specific skill requirements underlying an occupation. Therefore, we conduct additional tests to investigate how the supply of skills in the labor market moderate the value effects of augmenting AI. As discussed earlier, skill acquisition presents substantial challenges for firms when hiring frictions arise from skill mismatches and skill scarcity (McGuinness et al., 2018). The hiring frictions associated with specific skill requirements limit the realization of expected benefits from technologies. We posit that the supply of worker

skills outside the firm acts as a key factor in a firm's AI adoption. Access to the skill supply determines the extent to which firms can capture value from new AI innovations. Therefore, we expect that augmenting AI innovation creates more value for a firm when there is a sufficient supply of skills in the external labor market.

We employ three different measures of labor-market skill supply. First, we use industry-level skill diversity as a proxy for the cost of skill accumulation from the external labor markets. Second, we use industry-level supply of skills from (each category) AI-exposed occupations as a proxy for the availability of required skill sets provided by other AI-exposed occupations in the external labor markets. We expect that augmenting AI innovation (and especially scope-augmenting AI innovation) benefits firms more when they have better access to a sufficient supply of worker skills. Third, we use industry-level skill distance between (each category) AI-exposed occupations and other occupations to measure the skill proximity across occupations. When a firm's demand for novel skills increases, the firm may seek to hire across occupational boundaries. But when a firm's demand for existing (core) worker skills increases, hiring needs can be met within existing occupational boundaries. Based on this reasoning, we predict that core-augmenting AI (scope-augmenting AI) is more beneficial (not more beneficial) to firm value when there is a high skill proximity (i.e., small skill distance) between an exposed occupation and other occupations in the external labor markets. Similarly, we predict that scope-augmenting AI (core-augmenting AI) is more beneficial (not more beneficial) to firm value when there is a low skill proximity (i.e., large skill distance) between an exposed occupation and other occupations in the external labor market.

In Table 10, Panel A reports the results in subsamples of firms in high (above the sample median) and low industry-level skill diversity. This industry supply measure is calculated as one minus the Herfindahl-Hirschman Index (HHI) of skill concentration (measured by summing the squares of the employment shares at the skill level in each 2-digit SIC industry). We find significant positive value effects from all three categories of augmenting AI in Columns (1)-(3) but not in Columns (4)-(6), suggesting that firms benefit more from a diverse skill supply in external markets. In Panel B, Columns (1) through (3) report the results for the subsample of firms that have a high level of skill supply among AI-exposed occupations. Consistent with the results in Panel A, the positive coefficients in Columns (1)-(3) suggest that firms gain more value from labor-augmenting AI patents when the cost of skill accumulation is low in the external labor market.

Panel C reports the results in subsamples of firms with high (above the sample median) and low industry-level skill distance between AI-exposed occupations and others. This industry-level skill supply measure is calculated as the employment-share-weighted average of the occupational skill distance across (each category) AI-exposed occupations. For each occupation pair, skill distance is measured as the number of skills commonly required by the two (i.e., fewer common skills refer to a large skill distance). Then, for each AI-exposed occupation, we calculate the average distance across paired occupations weighted by their job transitions to the focal occupation. We find significant positive value effects of augmenting AI and scope-augmenting AI in Columns (1) and (2), but not (3). For core-augmenting AI, we find significant and positive value effects in Column (6) for the subsample of small skill distance. The results suggest that scope-augmenting AI, as it increases demand for new skills that were absent from the occupation, creates value when labor supply is from occupations with large skill distance (i.e., low skill proximity). However, core-augmenting AI, as it increases demand for existing skills that were mainly from the exposed occupation, creates value when labor supply is from occupations with small skill distance (i.e., high skill proximity).

Overall, the evidence in Table 10 shows that the direct supply and diversity of skills reduce firms' external hiring costs and create more value for firms that innovate labor-augmenting AI. In addition, scope and core augmenting AI may be subject to different hiring frictions as their demand for skills is required by different groups of occupations. Skill accumulation from occupations that share proximate (non-proximate) skill requirements in the external labor market can critically influence the amount of value a firm obtains from core-augmenting (scope-augmenting) AI innovation.

Lastly, we investigate the heterogeneous effects of displacing AI innovations. We focus on three potential moderators of cost-savings benefits. First, employees in states with less generous unemployment insurance (UI) benefits face more severe unemployment risk and thus demand a higher wage premium as compensation (Topel, 1984; Agrawal and Matsa, 2013). Second, a firm stands to gain more cost savings from displacing innovations when it currently has a greater number of employees in occupations exposed to displacement. Third, a firm should be able to save more costs when it has been paying relatively higher wages to its displaceable workers.

Table 11 reports the results of 2SLS regressions that test for these moderator effects. Columns (1) and (2) correspond to subsamples of firms headquartered in states with high and low UI benefits.

The state-level unemployment insurance data are obtained from the U.S. Department of Labor.³⁵ We measure unemployment insurance benefits as the product of average weekly benefits and total weeks compensated. Columns (3) and (4) consider subsamples of firms with high and low displaceable employment, where displaceable employment is measured as the firm’s total employees who work in occupations exposed to displacing AI innovation. In Columns (5) and (6), we consider subsamples of firms with high and low displaceable wages, measured as the firm’s total wages that are paid to workers in occupations exposed to displacing AI innovation. We find significant positive coefficients for *Log displacing patents* in Columns (2), (3), and (5), but not in the other columns. Hence, consistent with our predictions, firms with weaker in-state unemployment protection, more displaceable workers, and higher displaceable wages stand to gain more value from pursuing labor-displacing AI technologies.

6. Conclusion

The rapid pace of innovation in the field of AI has led to growing interest among academics, practitioners, and policymakers in understanding the possible benefits and consequences of new technologies that can process vast amounts of information, perform varied cognitive tasks, and engage with human users. While increasingly powerful AI technologies offer the promise of improving firms’ productivity within virtually every industrial sector, questions have arisen about the effects that AI will have on labor markets. Will AI complement human workers, thus increasing labor productivity and firms’ demand for new employees and new skills with which to leverage the new technology? Or, will AI be able to match and surpass humans in cognitive workplace tasks, leading to skill obsolescence, labor displacement, and technological unemployment? These questions have been difficult to answer due to the lack of large-scale data on the various types of AI innovation and the impact that these innovations have on the movements of individual workers to and from occupations and firms.

In this paper, we apply generative AI and LLMs to over five million patent filing texts from 2007-2023 to identify and distinguish different types of AI innovation. Using detailed data on worker flows at the occupation-by-firm level, we study the causal impact of these different AI innovation types on innovating firms’ employment growth and skill acquisition. Our work

³⁵ The data source is <https://oui.doleta.gov/unemploy/data.asp>

contributes to research on the labor-market impacts of AI along several dimensions. First, we introduce a new typology that facilitates the systematic study of AI in seven distinct areas of human-like cognition: language, perception, motor, engagement, decision-making, learning, and creativity. Second, by accounting for occupation-specific exposures to the seven AI types, we are able to provide causal evidence that is relevant to the recent debate about whether AI is likely to lead to the augmentation or displacement of human workers. In particular, we document that AI innovations related to language, engagement, decision-making, learning, or creativity significantly increase employment growth, but AI innovations related to perception or motor control significantly reduce it. Third, data on individual workers' skill sets enable us to test whether certain sub-categories of augmenting AI innovation cause firms to accumulate "novel" skills, i.e., skills that are not already common among the firms' pre-existing workforces. We find that AI innovations tied to engagement, learning, or creativity are "scope augmenting": they lead innovating firms to ramp up their hiring of workers with novel skills.

Our study also provides novel evidence on the implications of AI innovation for firm value and operating efficiency. While we find that both augmenting and displacing AI innovations translate into higher valuations for innovating firms, the two categories have different effects on efficiency. Specifically, augmenting innovations boost firms' total factor productivity, while displacing innovations improve cost efficiency but not productivity. We also find that the value impact of AI innovation depends heavily on the labor market circumstances surrounding the innovating firm. For example, augmenting AI innovation yields more value to an innovator in the presence of lower external hiring costs, i.e., when the labor mobility and supply are high, industry unionization rates are low, or there exists an ample external supply of workers with desired skills. Displacing AI, by contrast, is more value-enhancing to an innovating firm when the potential cost savings from displacement are high, such as when in-state unemployment insurance coverage is lower or when the firm has more displaceable wage costs or a greater supply of displaceable workers. Overall, our findings are consistent with the view that many AI innovations can yield favorable effects for both innovating firms and their workforces, but these positive value effects might only materialize when the costs of labor adjustment are low.

While offering novel insights into how AI innovation affects innovating firms' employment, valuation, and efficiency, our study also opens up new avenues for future work to explore other channels through which AI innovation could have real and financial effects. For instance, aside

from using AI to enhance their own productivity and efficiency, an innovative firm could sell AI-powered products and services to customers, or it could license AI-based inventions for use by other firms. Product innovation in the AI field could lead to a diverse set of effects, including the concentration of market power in some industries and widespread diffusion of new AI tools in others. Future research into these issues will help provide additional insights into the broader economic effects of AI innovation.

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Appendix A. Validation of LLM-Based Patent Classifications

We employ an “LLM-as-a-judge” approach (Zheng, 2023; Gu, 2025; Li, 2025) in which several frontier AI models serve as judges of the correctness of responses from the *Qwen2.5-14b-Instruct* open-source LLM. The judge models we use include the following state-of-the-art, proprietary models: *Gemini 2.5 Pro* (released June 17, 2025 by Google DeepMind); *Claude Opus 4.1* (released August 5, 2025 by Anthropic); *GPT 4.1* (released April 14, 2025 by OpenAI); and *DeepSeek R1 0528* (released May 28, 2025 by DeepSeek). As of early August 2025, all four models rank at or near the top of the LMArena leaderboard (<https://lmarena.ai/leaderboard>) from among 234 ranked models in terms of text-based generative AI performance.

We start by assessing the quality of basic inferences made by *Qwen2.5-14b-Instruct* about whether patent applications are AI or non-AI. For each year in the 2007-2023 sample period, we draw a random sample of 500 patent applications, resulting in a total sample of 8,500 applications. For each sampled application, we prompt *Qwen2.5-14b-Instruct* with “*Is the following invention directly related to artificial intelligence? Respond with YES or NO and give an explanation. [patent abstract text follows]*”. We then feed each response given by *Qwen2.5-14b-Instruct* to each of the four judge models using the following prompt: “*You will be presented with a question about an invention and a response to the question by an LLM. Please just state whether the response by the LLM is CORRECT or INCORRECT. [The prompt given to Qwen2.5-14b-Instruct and the resulting response text follow]*”.

For inferences about the seven individual AI types, we assess inference quality by drawing yearly random samples of 500 patent applications (2007-2023) from among those that were previously labeled by *Qwen2.5-14b-Instruct* as being AI-related. Then, for each application and each AI type ($8,500 \times 7 = 59,500$ inferences in total), we prompt *Qwen2.5-14b-Instruct* for a yes/no answer and an explanation of why the patent application does or does not belong to the type in question. Each response from *Qwen2.5-14b-Instruct* is then provided to each of the four judge models in conjunction with a prompt identical to the one used for the basic AI/non-AI categorizations.

From the assessments given by the four judge models, we calculate the accuracy of *Qwen2.5-14b-Instruct*’s overall AI/non-AI inferences as well as the accuracy of its inferences about each of the seven AI types. As seen in Table A.1, the percentage of AI/non-AI inferences judged to be correct ranges from 96% to 98.2%, depending on which judge model is being applied. For inferences about whether patent applications belong to each of the seven AI types, the percentage of inferences judged to be correct ranges from a low of 77.2% (engagement-based AI) to a high of 98.8% (creativity-based AI).

Table A.1. Using “LLM-as-a-Judge” to Evaluate the Correctness of Patent Classifications

This table shows the results of using several frontier AI models to assess the correctness of inferences made by the *Qwen2.5-14b-Instruct* LLM (Alibaba Cloud). Reported in the table are the percentages of sampled inferences from *Qwen2.5-14b-Instruct* that are deemed correct by the following four judge LLMs: *Gemini 2.5 Pro* (Google DeepMind), *Claude Opus 4.1* (Anthropic), *GPT 4.1* (OpenAI), and *DeepSeek R1 0528* (DeepSeek). The sample of inferences by *Qwen2.5-14b-Instruct* consists of a stratified random sample of 500 patent applications drawn from each year during 2007-2023. For the AI subtypes, inferences are based on a stratified random sample of 500 patent applications per year drawn from among those applications inferred by *Qwen2.5-14b-Instruct* to be directly related to AI.

AI Type	Judge Model			
	<i>Gemini 2.5 Pro</i>	<i>Claude Opus 4.1</i>	<i>GPT 4.1</i>	<i>DeepSeek R1 0528</i>
Any	97.7%	96.0%	97.0%	98.2%
Language	95.6%	92.5%	97.1%	97.0%
Perception	92.2%	82.6%	93.6%	95.7%
Motor control	96.6%	95.9%	97.9%	97.9%
Engagement	90.5%	77.2%	87.7%	91.6%
Decision-making	89.5%	83.0%	91.8%	89.6%
Learning	89.2%	77.3%	84.0%	87.8%
Creativity	96.7%	94.0%	97.1%	98.8%

Appendix B. Examples of AI Innovation Types Identified by LLM Queries

Language

Query: *Is the following invention directly related to artificial intelligence that has language ability? Respond with YES or NO and give an explanation.*

Title: DISPLAY-BASED CONTEXTUAL NATURAL LANGUAGE PROCESSING

Patent Application Number: 20220246139

Patent Applicant: Amazon Technologies Inc.

Abstract: Multi-modal natural language processing systems are provided. Some systems are context-aware systems that use multi-modal data to improve the accuracy of natural language understanding as it is applied to spoken language input. Machine learning architectures are provided that jointly model spoken language input (utterances) and information displayed on a visual display (on-screen information). Such machine learning architectures can improve upon, and solve problems inherent in, existing spoken language understanding systems that operate in multi-modal contexts.

Explanation from LLM:

YES. The described invention is directly related to artificial intelligence with language ability because it involves multi-modal natural language processing systems. These systems use both spoken language input and visual information to enhance natural language understanding, which falls under AI technologies that process and understand human language. Additionally, the use of machine learning architectures to model these inputs indicates an advanced form of AI capable of learning from various data types to improve its performance in understanding and processing natural language in multi-modal contexts.

Title: VIDEO ANALYSIS USING A DEEP FUSION REASONING ENGINE (DFRE)

Patent Application Number: 20220351521

Patent Applicant: Cisco Technology Inc.

Abstract: In one embodiment, a video analysis service receives video data captured by one or more cameras at a particular location. The service applies a neural network-based model to portions of the video data, to identify objects within the video data. The service maps outputs of the neural network-based model to symbols using a conceptual space. The outputs of the model comprise the identified objects. The service applies a symbolic reasoning engine to the symbols, to generate an alert. The service sends the alert to a user interface in conjunction with the video data.

Explanation from LLM:

NO. While the described invention involves elements of artificial intelligence, such as neural networks for object identification, it does not explicitly involve language abilities. The invention focuses on processing video data to identify objects and then uses a symbolic reasoning engine to generate alerts based on those identifications. Although mapping outputs to symbols could be seen

as a form of representation that might involve some level of abstraction or conceptual understanding, it does not necessarily imply language ability. Language ability typically involves more complex interactions such as understanding, generating, and interpreting natural language text or speech, which is not evident in this description.

Perception

Query: Is the following invention directly related to artificial intelligence that has perceptual ability? Respond with YES or NO and give an explanation.

Title: OFFLINE TRACKING SYSTEM FOR AUTONOMOUS VEHICLE CONTROL SYSTEMS

Patent Application Number: 20230294736

Patent Applicant: GM Cruise Holdings LLC

Abstract: Disclosed are systems, apparatuses, methods, and computer-readable media to autonomous driving vehicles and, in particular, for tracking objects in an environment that an autonomous vehicle (AV) is navigating. A method includes receiving environment data from at least one sensor in an AV control system mounted to the AV, the environment data including online tracking data that identifies at least one object within the environment data that is recorded at drive time; annotating the at least one object from the environment data that are incorrectly identified by the AV control system; executing an offline tracking engine to generate offline tracking data that tracks the objects over time in the environment data; and identifying safety gaps between the online tracking data and the offline tracking data.

Explanation from LLM:

YES. The invention described involves the use of sensors and tracking engines to identify and track objects in the environment of an autonomous vehicle (AV). This process inherently requires some form of perception and understanding of the environment, which are key aspects of artificial intelligence. The system uses both online and offline tracking mechanisms to ensure accurate identification of objects, indicating a level of cognitive processing and learning. The annotation step also suggests an improvement loop where the AI learns from its mistakes, enhancing its perceptual abilities over time. Therefore, this invention is indeed related to artificial intelligence with perceptual capabilities.

Title: IMAGE PROCESSING APPARATUS, IMAGE PROCESSING METHOD, AND STORAGE MEDIUM

Patent Application Number: 20220172470

Patent Applicant: Canon Inc.

Abstract: An image processing apparatus includes: a first designation unit configured to designate a first image group; a second designation unit configured to designate a second image group; an analysis unit configured to analyze each image included in the first image group and the second

image group; a determination unit configured to determine a reference, based on an analysis result of the second image group; and a selection unit configured to select an image from the first image group, based on the reference and an analysis result of the first image group.

Explanation from LLM:

NO. The described invention is related to image processing, which can be a component of artificial intelligence (AI) systems, but it does not necessarily imply perceptual ability as typically understood in AI contexts. The invention outlines a process for analyzing groups of images and selecting an image based on certain criteria, which could be part of a broader AI system. However, perceptual ability usually refers to more advanced cognitive processes such as recognizing objects, understanding scenes, or interpreting context, which go beyond mere analysis and selection based on predefined rules or references. This system seems to rely on pre-programmed functions rather than learning from experience or developing its own understanding of visual content, which would be necessary for true perceptual abilities in AI.

Motor Control

Query: *Is the following invention directly related to artificial intelligence that has motor skills? Respond with YES or NO and give an explanation.*

Title: EMERGENCY STOPPING FOR AUTONOMOUS COMMERCIAL VEHICLES

Patent Application Number: 20180052463

Patent Applicant: Omnitracs LLC

Abstract: The present disclosure generally relates to autonomous commercial vehicles. In one aspect, the disclosure provides a method for controlling a commercial highway vehicle. The method includes detecting a failure of a first component based on a first signal from a first sensor. The method also includes classifying, by an automated driving system on the vehicle, a severity of the component failure. The method further includes determining to stop the vehicle if the severity exceeds a threshold severity level. The method also includes determining an emergency stopping distance based on the severity and a current momentum of the vehicle. The method further includes determining a stopping location within the emergency stopping distance. The method also includes stopping the vehicle at the stopping location. The present disclosure also provides an autonomous commercial vehicle and an emergency control system for performing the method.

Explanation from LLM:

YES. The invention described involves an autonomous commercial vehicle that can detect and classify failures in its components, determine appropriate actions based on the severity of those failures, and perform actions such as stopping the vehicle safely. This process involves the use of sensors to gather data (like the status of components), automated systems to analyze this data and make decisions (such as whether a failure is severe enough to warrant stopping), and then executing those decisions (like stopping the vehicle). These functionalities require advanced computational capabilities and decision-making processes, which are hallmarks of artificial intelligence. Additionally, the act of stopping the vehicle at a determined location involves motor

skills or precise control over the vehicle's movement, demonstrating AI integration with physical action.

Title: METHOD AND SYSTEM FOR REAL-TIME AND OFFLINE DE-IDENTIFICATION OF FACIAL REGIONS FROM REGULAR AND OCCLUDED COLOR VIDEO STREAMS OBTAINED DURING DIAGNOSTIC MEDICAL PROCEDURES

Patent Application Number: 20210209734

Patent Applicant: GE Precision Healthcare LLC

Abstract: Systems and techniques that facilitate real-time and/or offline de-identification of facial regions from regular and/or occluded color video streams obtained during diagnostic medical procedures are provided. A detection component can generate a bounding box substantially around a person in a frame of a video stream, can generate a heatmap showing key points or anatomical masks of the person based on the bounding box, and can localize a face or facial region of the person based on the key points or anatomical masks. An anonymization component can anonymize pixels in the frame that correspond to the face or facial region. A tracking component can track the face or facial region in a subsequent frame based on a structural similarity index between the frame and the subsequent frame being above a threshold. If the structural similarity index between the frame and the subsequent frame is above the threshold, the tracking component can track the face or facial region in the subsequent frame without having the detection component generate a bounding box or a heatmap in the subsequent frame, and the anonymization component can anonymize pixels in the subsequent frame corresponding to the tracked face or facial region.

Explanation from LLM:

NO. The described invention is not directly related to artificial intelligence with motor skills. The system and techniques mentioned focus on de-identifying facial regions from video streams, which involves computer vision and image processing tasks such as object detection (face localization), heatmap generation for key points, and pixel anonymization. While these tasks require sophisticated AI algorithms, they do not involve motor skills, which typically refer to the ability to control physical movements or manipulate objects in the environment. This invention is more about privacy protection and data security in medical imaging rather than interacting with or controlling physical devices or systems.

Engagement

Query: *Is the following invention directly related to artificial intelligence that has the ability to engage with users? Respond with YES or NO and give an explanation.*

Title: INITIALIZING A CONVERSATION WITH AN AUTOMATED AGENT VIA SELECTABLE GRAPHICAL ELEMENT

Patent Application Number: 20230336505

Patent Applicant: Google LLC

Abstract: Methods, apparatus, systems, and computer-readable media are provided for invoking an agent module in an automated assistant application in response to user selection of a selectable element presented at a graphical user interface rendered by a non-automated assistant application. The invoked agent module can be associated with other content rendered in the non-automated assistant graphical user interface, and can optionally be invoked with values that are based on user interactions via the non-automated assistant application. Responsive content can be received from the agent module in response to the invocation, and corresponding content provided by the automated assistant application via an automated assistant interface. In these and other manners, selection of the selectable element causes transition from a non-conversational interface, to a conversational automated assistant interface—where an agent (relevant to content in the non-conversational interface) is invoked in the automated assistant interface.

Explanation from LLM:

YES. The described invention is indeed directly related to artificial intelligence as it involves engaging with users through an automated assistant application. This system allows for the transition from a non-conversational interface to a conversational one where an AI agent can be invoked based on user interaction within another application. The agent module's ability to provide relevant responses based on user interactions and the context of the interface indicates the use of AI technologies such as natural language processing (NLP), machine learning, and possibly more advanced AI techniques for understanding context and generating appropriate responses. Therefore, this invention is clearly designed to facilitate user engagement through intelligent, interactive means, which is a hallmark of AI-based inventions.

Title: DYNAMICALLY RECONFIGURABLE INSURANCE PRODUCT

Patent Application Number: 20210258756

Patent Applicant: Accenture Global Solutions Ltd.

Abstract: A location-based risk alerting system is disclosed. When information regarding a risk assessment situation is received, the location and time attributes of the risk assessment situation are extracted, a risk score is calculated and a risk alert is generated to include the attributes and the risk score upon validating the received information for risk. The risk alert is transmitted to client devices of subscribers where a user-specific risk score is calculated at each client device based on a comparison of the distance of the client device from the location attribute and the time at which the risk alert is received with the time attribute. The risk alert is displayed at the client device based on the user-specific risk score and information regarding the user is shared with trusted authorities upon receiving user consent.

Explanation from LLM:

NO. While this invention involves sophisticated processing and decision-making, it does not necessarily imply direct engagement with users in the sense typical of AI systems designed for interaction, such as chatbots or virtual assistants. Instead, it primarily focuses on automated processes for generating and distributing alerts based on predefined criteria (location, time, risk assessment). Although there is some user interaction implied in the context of receiving alerts and potentially sharing information with authorities, the core functionality described does not

inherently require engaging with users through natural language processing, learning from interactions, or other AI-driven conversational techniques.

Decision-making:

Query: Is the following invention directly related to artificial intelligence that has decision-making ability? Respond with YES or NO and give an explanation.

Title: SYSTEM AND METHOD FOR TRACKING EGRESS TIMES FROM A PARKING FACILITY AND PROVIDING ACTION RECOMMENDATIONS

Patent Application Number: 20230141416

Patent Applicant: Honeywell International Inc.

Abstract: A method includes receiving vehicle data for a plurality of vehicles in a parking facility over time, the vehicle data including a time t1 indicating when each of the vehicles leaves its corresponding parking spot in the parking facility, and a time t2 indicating when each of the plurality of vehicles reaches an exit of the parking facility. An egress time is determined for each of the plurality of vehicles based at least in part on the corresponding times t1 and t2. A timestamp is assigned to each of the egress times, resulting in timestamped egress times. A current egress time of the parking facility is estimated based at least in part on the timestamped egress times. An action recommendation is based at least in part on the current egress time of the parking facility and is outputted for subsequent viewing by the driver of the particular vehicle.

Explanation from LLM:

YES. This invention involves collecting and processing data from vehicles within a parking facility to estimate the egress (exit) time of the facility. The system uses this data to make recommendations to drivers, which implies it has some form of decision-making capability. Specifically, the system analyzes historical data (times t1 and t2) to predict current conditions (current egress time) and then uses these predictions to suggest actions to drivers. This process involves pattern recognition, prediction, and decision-making, all of which are elements of artificial intelligence.

Title: METHOD AND SYSTEM FOR NAVIGATING AT A CLIENT DEVICE SELECTED FEATURES ON A NON-DYNAMIC IMAGE PAGE FROM AN ELASTIC VOICE CLOUD SERVER IN COMMUNICATION WITH A THIRD-PARTY SEARCH SERVICE

Patent Application Number: 20210099768

Patent Applicant: Dish Network LLC

Abstract: Systems, and methods are described for navigating non-dynamic pages at a client device served up from voice search requests by selecting a feature on the image page by a user request for navigating the feature viewed on the image page wherein the feature selected has an appearance of a dynamic feature when served up on the display wherein the user request selects a location on the image page; identifying the feature by the voice cloud server by location data of a selected

location on the page; executing a feature request on a corresponding page which contains dynamic script at the voice cloud server to receive a linked page from a third-party search service; and removing dynamic scripts, at the voice cloud search server, of the linked page for serving up as an image page of the linked page at the client device.

Explanation from LLM:

NO. The described invention involves navigating through web pages using voice commands and ensuring that the resulting pages are displayed without dynamic content (such as JavaScript) for better performance or accessibility. While it uses AI in the form of voice recognition to interpret user requests, the core function is not about AI making decisions but rather about processing user commands and serving simplified versions of web pages based on those commands. The decision-making aspect here is relatively straightforward and does not involve complex reasoning or learning capabilities typical of advanced AI systems.

Learning:

Query: *Is the following invention directly related to artificial intelligence that has learning ability? Respond with YES or NO and give an explanation.*

Title: SYSTEMS AND METHODS FOR UTILIZING TIME SERIES AND NEURAL NETWORK MODELS TO DEPLOY AUTONOMOUS VEHICLES FOR 5G NETWORK COVERAGE GAPS

Patent Application Number: 20210297878

Patent Applicant: Verizon Patent and Licensing Inc.

Abstract: A device may receive historical usage data associated with a portion of a wireless network, and may train a model with the historical usage data to generate a trained model. The device may process data identifying a future time period, with the trained model, to forecast a traffic demand during the future time period, and may compare the traffic demand and a traffic capacity of the portion of the wireless network to determine whether the traffic demand is within a threshold of or exceeds the traffic capacity. The device may identify an autonomous vehicle to deploy for the portion of the wireless network when the traffic demand is within the threshold of or exceeds the traffic capacity, and may cause the autonomous vehicle to be deployed for the portion of the wireless network, wherein the autonomous vehicle is to provide a temporary wireless network.

Explanation from LLM:

YES. The described invention involves the use of a model that is trained on historical usage data from a wireless network, which indicates a form of machine learning where the system can analyze past patterns to make predictions about future traffic demands. This process of training a model on data and using it to make forecasts is a fundamental aspect of artificial intelligence (AI) with learning abilities. Furthermore, the system uses these predictions to make decisions about deploying resources (in this case, an autonomous vehicle providing a temporary wireless network),

which showcases an AI's ability to not only learn but also to take actions based on learned insights. Therefore, the invention is directly related to AI that has learning capabilities.

Title: GEORADIO

Patent Application Number: 20220067080

Patent Applicant: Gotham Studios Inc.

Abstract: Systems, methods, and computer-program products are described for determining current location information which includes determining whether an object is moving by comparing the current location information with previous location information. Future location information is calculated and a point of interest is filtered using the future location information. A media asset is then matched to the filtered point of interest and a current score is determined for the matched media asset. Determining a current score includes determining a window in which the matched media asset is geographically relevant to the future location information, and updating the matched media asset score when the future location information changes. An ordered playlist is dynamically generated according to the future location information by repeatedly prioritizing and sequencing matched media assets according to the current score for each matched media asset, and matched media assets are played or displayed in the ordered playlist.

Explanation from LLM:

NO. While the system described involves sophisticated algorithms for predicting future locations and dynamically adjusting content based on predicted user movement, it does not explicitly describe a learning process where the system improves its performance over time through experience. The described system relies on predefined rules and calculations rather than a self-improving learning mechanism typical of AI systems with learning abilities. It uses current and future location information to filter points of interest and match media assets, but this is more about predictive analytics and dynamic content adaptation than learning from interactions or data feedback to improve its performance iteratively.

Creativity:

Query: *Is the following invention directly related to artificial intelligence that can generate content? Respond with YES or NO and give an explanation.*

Title: BODY FITTED ACCESSORY WITH PHYSICS SIMULATION

Patent Application Number: 20230074826

Patent Applicant: Snap Inc.

Abstract: Methods and systems are disclosed for performing operations comprising: receiving video that includes a depiction of a real-world object; generating a three-dimensional (3D) body mesh associated with the real-world object that tracks movement of the real-world object across frames of the video; obtaining an external mesh associated with an augmented reality (AR) element; determining that a first portion of the external mesh is associated with movement information of

the 3D body mesh; determining that a second portion of the external mesh is associated with movement information of an external force model; deforming the first and second portions of the external mesh separately based on movement information of the 3D body mesh and movement information of the external force model; and modifying the video to include a display of the AR element based on the deformed first and second portions of the external mesh.

Explanation from LLM:

YES. This invention involves the use of AI in generating and manipulating content, specifically augmented reality (AR) elements in videos. It uses machine learning techniques to track movements of real-world objects within a video and apply these movements to AR elements. The process of generating a 3D body mesh, determining how different parts of an AR element should move based on the movements of the real-world object and external forces, and then deforming and displaying these elements requires sophisticated algorithms that are typically powered by AI technologies. This demonstrates direct involvement with AI-generated content as it involves creating and integrating synthetic elements into real-world footage dynamically.

Title: AUTOMATED IMAGE ANALYSIS USING ARTIFICIAL INTELLIGENCE TECHNIQUES

Patent Application Number: 20230410476

Patent Applicant: Dell Products LP

Abstract: Methods, apparatus, and processor-readable storage media for automated image analysis using artificial intelligence techniques are provided herein. An example computer-implemented method includes obtaining image data associated with at least one event; characterizing a plurality of attributes of the image data by processing at least a portion of the image data using one or more artificial intelligence techniques and processing metadata associated with at least a portion of the image data using one or more rule-based techniques; generating a determination as to whether one or more portions of the image data have been modified based at least in part on the plurality of characterized attributes of the image data; and performing one or more automated actions, based at least in part on the determination, in connection with at least one user request related to the image data associated with the at least one event.

Explanation from LLM:

NO. The described invention is related to automated image analysis using artificial intelligence (AI) techniques but does not specifically involve AI content generation. The focus is on analyzing images, detecting modifications within them, and taking automated actions based on this analysis. While it uses AI techniques for processing image data and metadata, the primary function is not about generating new content such as text, images, or other forms of media from scratch, which would be characteristic of AI content generation systems.

Appendix C. Occupational Exposure to AI and Individual Workers' Skills

Table C.1. Occupations with High Exposure to Different AI Types Based on O*NET Skill Requirements

This table shows, for each type of AI innovation, the three 3-digit SOC occupations that have the highest degree of skill-based exposure to that AI type. An occupation's degree of exposure to an AI type is calculated as the occupation's weighted count of skill requirements in the O*NET database that map to the AI type, where the weights are the O*NET importance scores of skill requirements. The mapping between skill requirements and AI types is constructed as described in Section 3.3.2. The rightmost column shows, for each AI type, the three skill requirements that have the highest O*NET importance scores across the three highly-exposed occupations.

AI Type	Most-Exposed Occupation		2 nd -Most Exposed Occupation		3 rd -Most Exposed Occupation		Most Important O*NET Skill Requirements
	3-Digit SOC	Title	3-Digit SOC	Title	3-Digit SOC	Title	
Language	111	Top Executives	212	Religious Workers	431	Supervisors of Office and Administrative Support Workers	Reading Comprehension; Writing; Service Orientation
Perception	491	Supervisors of Installation, Maintenance, and Repair Workers	492	Electrical and Electronic Equipment Mechanics, Installers, and Repairers	493	Vehicle and Mobile Equipment Mechanics, Installers, and Repairers	Repairing; Troubleshooting; Management of Material Resources
Motor control	431	Supervisors of Office and Administrative Support Workers	491	Supervisors of Installation, Maintenance, and Repair Workers	492	Electrical and Electronic Equipment Mechanics, Installers, and Repairers	Monitoring; Reading Comprehension; Repairing
Engagement	451	Supervisors of Farming, Fishing, and Forestry Workers	491	Supervisors of Installation, Maintenance, and Repair Workers	535	Water Transportation Workers	Operations Monitoring; Management of Personnel Resources; Active Listening

(Continued)

Table C.1, Continued

AI Type	Most-Exposed Occupation		2 nd -Most Exposed Occupation		3 rd -Most Exposed Occupation		Most Important O*NET Skill Requirements
	3-Digit SOC	Title	3-Digit SOC	Title	3-Digit SOC	Title	
Decision-making	451	Supervisors of Farming, Fishing, and Forestry Workers	491	Supervisors of Installation, Maintenance, and Repair Workers	511	Supervisors of Production Workers	Coordination; Critical Thinking; Judgment and Decision Making
Learning	172	Engineers	451	Supervisors of Farming, Fishing, and Forestry Workers	491	Supervisors of Installation, Maintenance, and Repair Workers	Critical Thinking; Monitoring; Management of Personnel Resources
Creativity	111	Top Executives	172	Engineers	491	Supervisors of Installation, Maintenance, and Repair Workers	Complex Problem Solving; Critical Thinking; Speaking

Table C.2. Individual Workers' Skills

This table shows examples of skill sets held by individual workers undergoing job transitions within a firm or between a firm and its external labor market. Data on job transitions are drawn from Revelio Labs' inflow transitions and outflow transitions datasets. Data on skill sets are compiled from Revelio Labs' individual user skills dataset. Panel A shows examples of skill sets held by newly-hired workers, while Panel B shows examples of skill sets held by workers departing a job.

Panel A: Examples of skills held by newly-hired workers					
Employee ID	Firm	Year	SOC 3-Digit Occupation	Within-Firm Transition?	Employee's Skill Set
49262072	STARBUCKS CORP	2014	Supervisors of Sales Workers (411)	No	call centers, customer satisfaction, customer service, insurance, inventory management, leadership, management, merchandising, Microsoft excel, Microsoft office, Microsoft word, PowerPoint, process improvement, retail, sales, team building, team leadership, time management, training, visual merchandising
49286590	NATIONAL INSTRUMENTS CORP	2015	Engineers (172)	Yes	analog, application-specific integrated circuits (ASIC), automation, C, C++, data acquisition, debugging, electrical engineering, electronics, embedded software, embedded systems, engineering, field-programmable gate arrays (FPGA), integrated circuits (IC), LabVIEW, manufacturing, MATLAB, PCB design, product development, product management, project management, semiconductors, soc, software tests, system design, test automation, test engineering, test equipment, testing
49267198	ROYALITE PETROLEUM CO INC	2014	Media and Communication Workers (273)	No	advertising, business development, business management, business strategy, creativity, customer relationship management (CRM), digital marketing, e-commerce, leadership, management, marketing, marketing strategy, Microsoft office, mobile devices, negotiation, online advertising, online marketing, product management, project management, public speaking, research, social media, social networks, Spanish, strategic planning, strategy, team leadership, teamwork, telecommunications, training

(Continued)

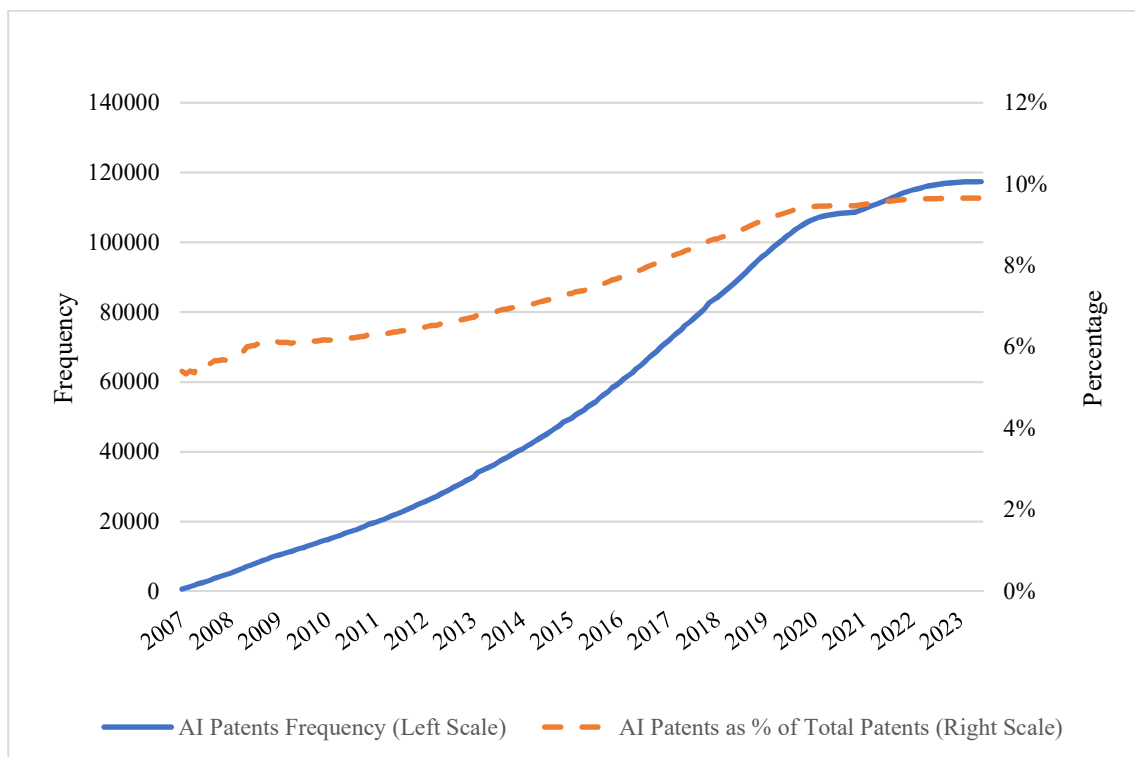
Table C.2, Continued

Panel B: Examples of skills held by departing workers					
Employee ID	Firm	Year	SOC 3-Digit Occupation	Within-Firm Transition?	Employee's Skill Set
49282976	VMWARE INC-CL A	2010	Computer Occupations (151)	Yes	cisco technologies, cloud computing, data center, disaster recovery, information technology infrastructure library, integration, IT service management, Linux, network security, networking, requirements analysis, servers, software development, storage area networks, structured query language, TCP/IP, UNIX, virtualization, VMware, windows server
49250337	STARBUCKS CORP	2016	Supervisors of Food Preparation and Serving Workers (351)	No	catering, cooking, culinary skills, customer service, event planning, food, food & beverage, food safety, food service, hospitality, hospitality management, inventory management, leadership, management, menu development, Microsoft excel, Microsoft office, Microsoft word, PowerPoint, process improvement, project management, research, restaurant management, restaurants, sales, team building, time management, training
49285556	WALMART INC	2020	Other Transportation Workers (536)	No	cashiering, customer satisfaction, customer service, data entry, inventory control, inventory management, leadership, loss prevention, management, merchandising, Microsoft excel, Microsoft office, Microsoft word, PowerPoint, retail, retail sales, sales, team building, teamwork, time management

Figure 1. Growth in AI Innovation Activity Over Time

This figure illustrates the time series of AI patenting activity by firms in CRSP/Compustat during 2007-2023. Patent grants are counted as of their application filing dates. AI patents and their functional categories are identified as described in Section 3.2.1 by applying generative LLM techniques to patent text data from the USPTO Bulk Data Storage System (BDSS). In Panel A, the solid graph corresponds to the cumulative monthly frequency of AI patents (left scale), while the dashed graph shows the cumulative monthly frequency of AI patents as a percentage of the cumulative monthly frequency of all patents (right scale). Panel B shows the time series of cumulative monthly patent frequencies for each of the seven AI types that constitute the functional typology outlined in Section 2.

Panel A. AI innovation and overall innovation



Panel B. AI innovation, by type

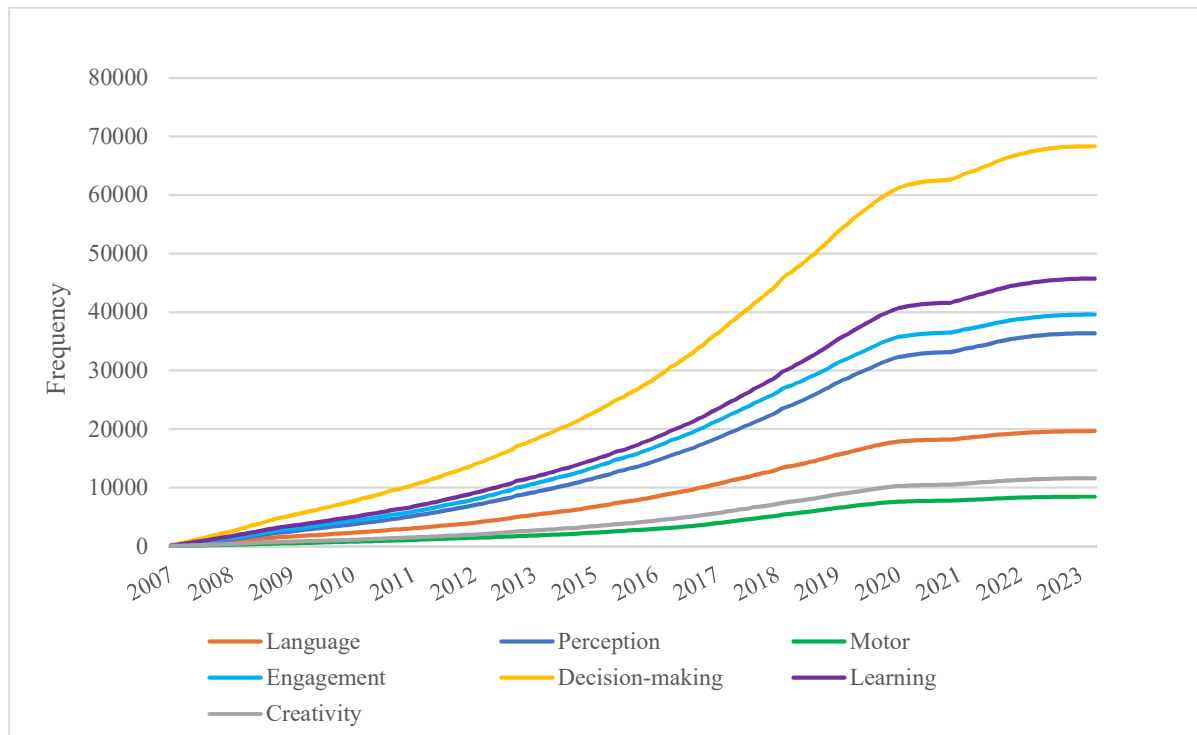
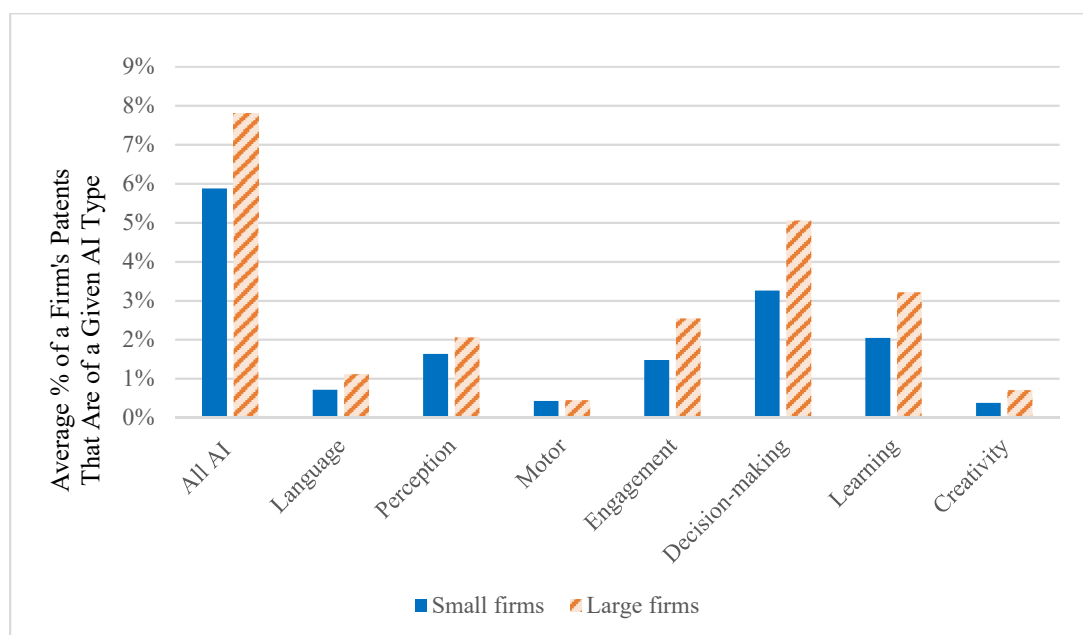


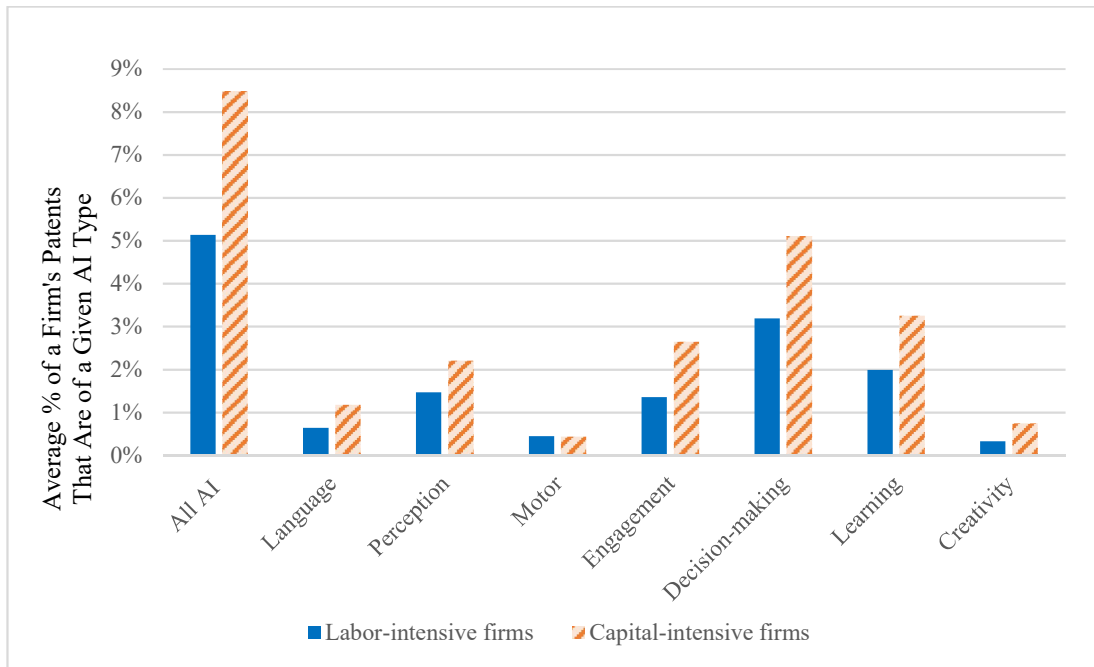
Figure 2. AI Innovation Activity and Firm Characteristics

This figure shows the extent of different types of AI patenting by firms in CRSP/Compustat during 2007-2023. AI types are defined according to the functional typology outlined in Section 2. AI patents and different types of AI patents are identified as described in Section 3.2.1 by applying generative LLM techniques to patent text data from the USPTO Bulk Data Storage System (BDSS). Panel A reports, by AI type and for small and large firms, the average percentage of a firm's total patents during a year that belong to the category. A firm is large (small) if its log-transformed total assets is above (below) the sample median during the year. Panel B reports, by AI type and for labor-intensive and capital-intensive firms, the average percentage of a firm's total patents during a year that are of that type. A firm is defined as labor-intensive (capital-intensive) if the firm's total employment scaled by total assets is above (below) the sample median during the year. Firm-level employment data are obtained from the Revelio Labs database. Panel C reports, by AI type and for R&D-intensive and non-R&D-intensive firms, the average percentage of a firm's total patents during a year that are of that type. A firm is R&D-intensive (non-R&D-intensive) if the firm's R&D expenditure scaled by total assets is above (below) the sample median during the year.

Panel A. AI innovation activity by large and small firms



Panel B. AI innovation activity by labor-intensive and capital-intensive firms



Panel C. AI innovation activity by R&D-intensive and non-R&D-intensive firms

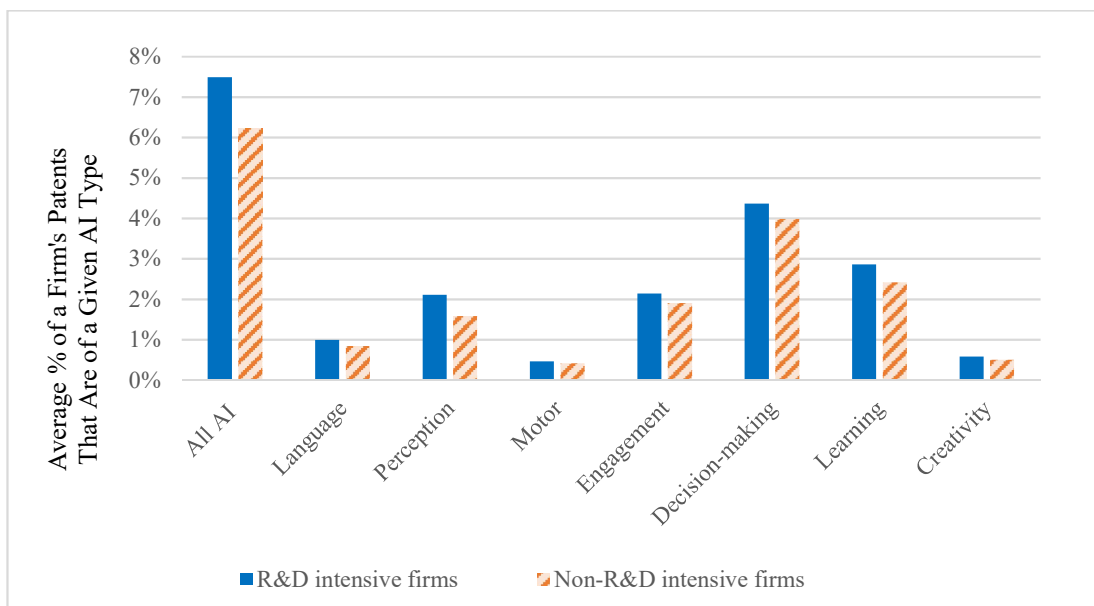


Figure 3: AI Innovation Activity and Occupational Employment Changes

This figure shows a heatmap depicting how different types of AI innovation activity relate to subsequent changes in occupational employment at innovating firms. The heatmap is based on observed employment changes at the firm-year-occupation level (occupations are disaggregated to the 3-digit SOC level). To construct the heatmap, we first identify cases in which a firm has a predominant AI innovation type in year t , i.e., a type that accounts for the largest fraction and more than one-third of all the firm's AI patents during the year. We then sort observations into groups according to the following dimensions: occupation, year, firm-size quartile (total assets), and predominant AI type. For each observation, we calculate an employment growth score based on the quartile ranking of the observation's year $t+1$ employment change within the group (1 = bottom quartile, 4 = top quartile). We then construct the cells of the heatmap by standardizing scores to lie between -1 and 1 and averaging within pairs formed by AI types and 2-digit SOC occupational groups. Darker red in a cell indicates that the relevant AI type is associated with more positive employment changes in an occupational group, while darker blue indicates more negative employment changes.

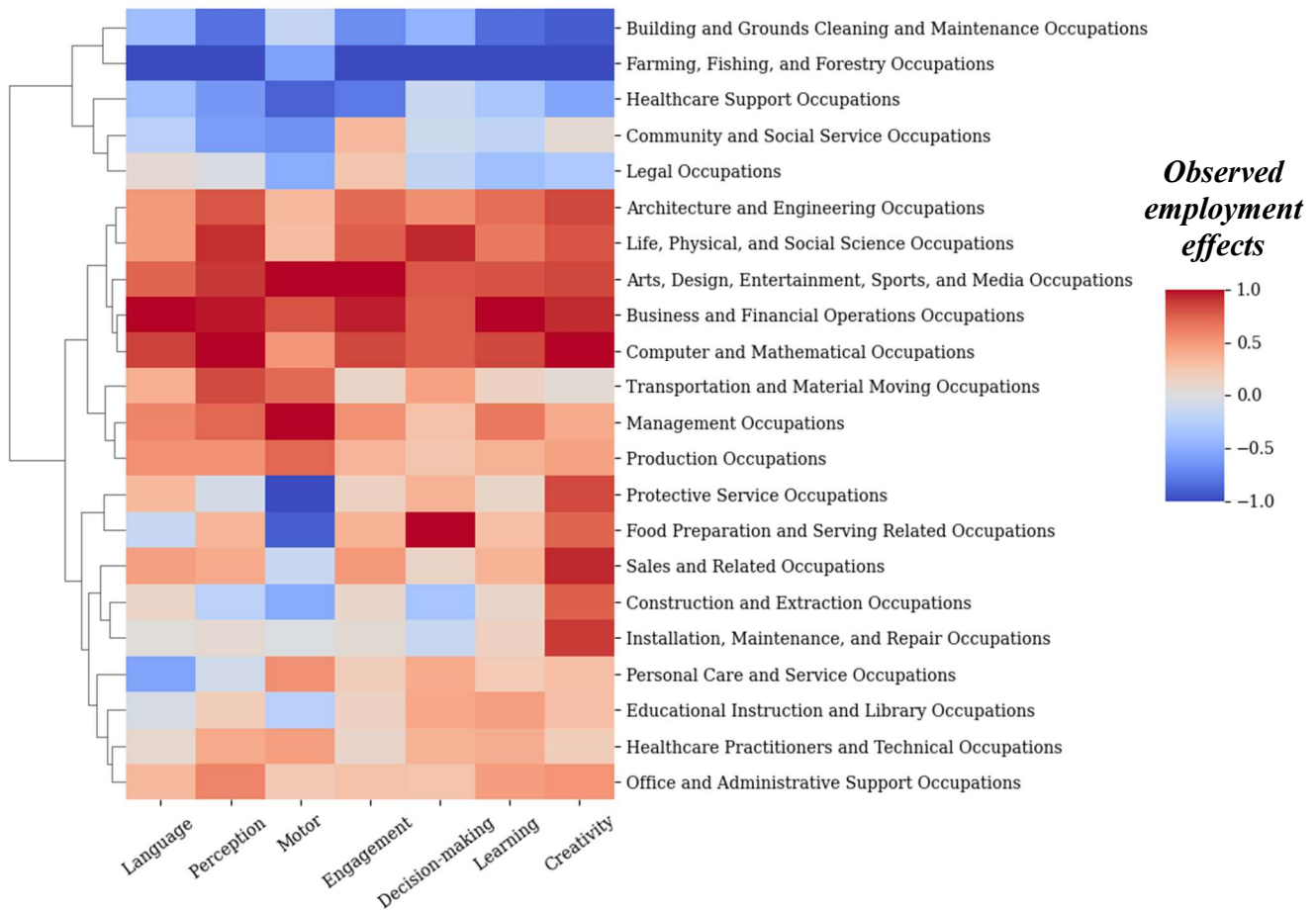


Figure 4. Categorization of AI Types

This figure shows a grouping of different AI innovation types into broad categories according to their estimated effects on occupational employment. Section 2 outlines our functional typology consisting of seven different AI types. As detailed in Section 3.2.1, we identify different AI types in the patent data by applying generative LLM techniques to patent documents from the USPTO Bulk Data Storage System (BDSS). Based on the estimated employment effects in Tables 4 and 5, we categorize a particular AI patent type as *Augmenting*, *Scope augmenting*, *Core augmenting*, or *Displacing* if it impacts exposed occupations by increasing overall employment, increasing employment of workers with novel skills, increasing overall employment but not employment of workers with novel skills, or decreasing overall employment, respectively. In the figure, the heights of the colored bars indicate the frequencies of different AI patent types among CRSP/Compustat firms during 2007-2023.

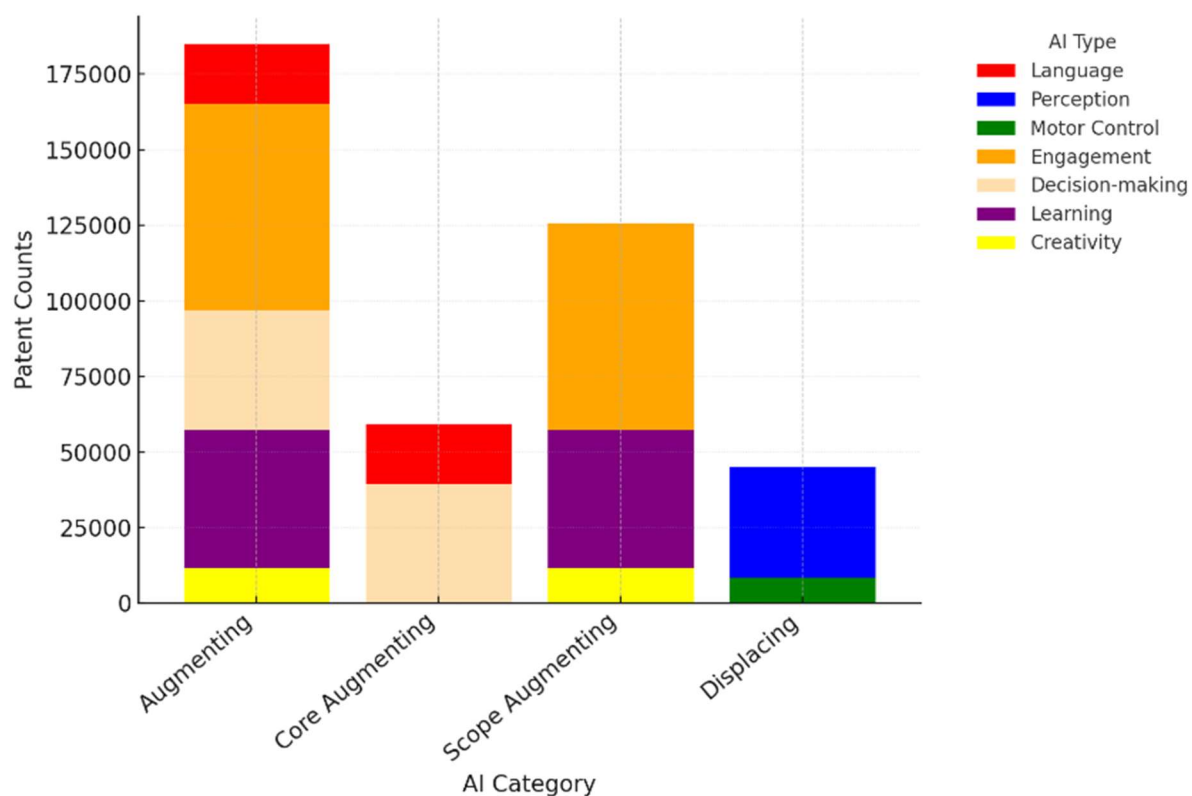


Table 1. A Functional Typology of Artificial Intelligence

This table summarizes an AI typology based on functional capabilities (see Section 2 for details). The first column in the table lists a set of single-term labels for the seven kinds of AI in our typology. The second column shows a sampling of descriptions from the psychology, cognitive science, and computer science literatures that, in conjunction with Newell's theory of cognition, define and characterize the seven AI types. The third column shows examples of real-world applications related to each AI type. Examples of real-world applications are collected from various public sources, including news media, research papers, and industry reports.

AI Type (Functional Capability)	Descriptions and Definitions from Psychology, Cognitive Science, and Computer Science	Examples of AI Applications
<i>Language</i>	Phonology and phonotactics, reading, spelling, lexis, morphosyntax, formulaic language, language comprehension, grammaticality, sentence production, and syntax (Carruthers, 2002; Kravchenko, 2007)	Large language models (LLMs); Dubbing for video localization; Voiceprint analysis; Real-time voice translation; Live call-center interpretation
<i>Perception</i>	Reception of stimuli, extraction and storage of perceptual symbols, and updating of perceptual systems (Barsalou, 1999; Wang, Hahm, and Hammer, 2022)	Computer vision systems; Autonomous vehicle systems; Diagnostic medical imaging; Augmented reality (AR) and Virtual reality (VR)
<i>Motor control</i>	Control, execution, and adjustment of goal-directed actions based on visual, proprioceptive, and kinesthetic feedback (Grush, 2004; Rosenbaum, 2009)	Autonomous ground vehicles (AGVs), drones, and aircraft; AI-powered humanoid robots; Mobile vehicle swarms; Servo motor control and monitoring; Smart home sensors
<i>Engagement</i>	Selective engagement with the environment based on cognitive motivation and emotion (Simon, 1967; Deci and Ryan, 2000)	Siri; Cortana; Alexa; Chat GPT; customer service AI; Videogame NPCs; Smart home devices; Self-driving delivery robots

(Continued)

Table 1, Continued

<i>Decision-making</i>	Setting plans through a reasoning function and selecting subjectively rational choices based on information about uncertainties and preferences (Pomerol, 1997; Fox, Cooper, and Glasspool, 2013)	Automated credit decision; AI resume screening; Autonomous vehicle navigation
<i>Learning</i>	Acquiring skills or experience in one situation and utilizing them in another situation (Greeno, Collins, and Resnick, 1996; Silver et al., 2021)	AlphaFold protein structure prediction; Personalized recommendation engines; Machine learning systems for healthcare diagnostics; Fraud and anomaly detection systems
<i>Creativity</i>	Actively and consciously pursuing an innovative solution or a new idea with the novelty derived from imagination (Pelaprat and Cole, 2011; Stokes, 2014)	ChatGPT; Adobe Firefly art generator; OpenAI Sora video generator; Robotic painters and sculptors

Table 2. AI Innovation Activity by Industry Sector and IPC Class

This table summarizes, for each type of AI innovation, the extent of patenting by CRSP/Compustat firms across different industrial sectors (Panel A) and across the 10 most frequently represented IPC classes (Panel B) during 2007-2023. Patent grants are counted as of the application filing date. Section 2 outlines our functional typology consisting of seven different AI types. As detailed in Section 3.2.1, we identify different AI types in the patent data by applying generative LLM techniques to patent documents from the USPTO Bulk Data Storage System (BDSS). If a patent is jointly filed by multiple firms, we report the industry sector of its largest (by total assets) innovating firm.

Panel A: Industry distribution of AI patent grants

Description	SIC (1-digit)	All AI	Language	Perception	Motor control	Engagement	Decision- making	Learning	Creativity
Manufacturing	3	47,440	4,106	17,256	6,042	11,625	26,494	15,389	2,332
Transport; storage and communication	7	46,263	11,873	12,291	1,236	20,308	27,533	20,735	7,088
Electricity; gas and water supply	4	8,653	1,770	2,530	336	3,398	5,011	3,461	1,073
Wholesale and retail trade	6	4,516	751	1,086	47	1,636	2,978	1,888	413
Community; social and personal services	9	3,732	214	1,254	254	551	2,197	1,414	120
Mining and quarrying	2	2,453	110	719	307	370	1,410	894	109
Construction	5	2,029	440	581	157	928	1,182	855	249
Financial; insurance; real estate and business services	8	1,483	359	473	25	655	1,021	761	198
Agriculture; hunting; forestry and fishing	1	601	21	119	30	49	384	252	15
Others	0	33	0	11	4	2	11	6	0

(Continued)

Table 2, Continued

Panel B: Distribution of AI patents across the 10 most frequently represented IPC classes									
Description	IPC Class	All AI	Language	Perception	Motor control	Engagement	Decision-making	Learning	Creativity
Computing, Calculating, or Counting	G06	75,879	14,593	24,478	3,055	29,380	43,827	32,698	9,800
Electric Communication Technique	H04	31,831	4,306	8,629	771	10,276	17,840	10,410	2,219
Musical Instruments; Acoustics	G10	7,561	5,235	3,807	98	5,460	3,962	3,954	1,765
Measuring; Testing	G01	6,412	230	2,256	823	981	3,920	2,102	94
Medical or Veterinary Science; Hygiene	A61	6,066	148	2,243	943	852	3,427	1,944	67
Controlling; Regulating	G05	5,372	205	2,265	2,502	1,224	4,543	2,167	75
Vehicles in General	B60	5,041	171	2,678	2,062	1,122	4,342	1,702	32
Signaling	G08	3,943	172	1,828	849	914	3,232	1,224	34
Educating; Cryptography; Display; Advertising; Sales	G09	2,089	376	949	149	1,073	1,007	733	244
Hand tools; Portable power-driven tools; Manipulators	B25	1,258	80	578	1,071	428	801	390	23

Table 3. Summary Statistics: Occupational Employment and Firm Characteristics

This table reports sample summary statistics. Panel A shows statistics for yearly occupation-by-firm observations during 2007-2023. *Employment growth* (*Employment growth (novel skills)*) is the within-firm occupation-level change in the total number of employees (number of employees with novel skills), multiplied by 10,000 and divided by the firm's prior-year total number of employees. *Exposed to type- k AI* (where type k is one of the seven AI types) is the fraction of the firm's total employment across 3-digit SOC occupations that have at least four important skill requirements (O*NET importance score ≥ 3) that map to AI type k . Patent types are categorized as *Augmenting*, *Scope augmenting*, *Core augmenting*, or *Displacing* according to how they affect exposed occupational employment in Tables 4 and 5 (see Figure 4). *ROA* is income before extraordinary items divided by total assets, both measured in the prior year. *Market-to-book* is the market-to-book ratio in the prior year. *R&D* is research and development expense in the prior year; missing values are imputed as zero. *TFP growth* is the yearly growth rate of total factor productivity (TFP), where TFP is the estimated residual from a firm-year linear production function model with capital and labor inputs. $\Delta SG\&A$ is the yearly change in the natural logarithm of (one plus) selling, general, and administrative expenditures (Compustat item *XSGA*). $\Delta Operating\ expense$ is the yearly change in the natural logarithm of (one plus) operating expenses (Compustat item *XOPR*). $\Delta Tobin's\ Q$ is measured as the yearly change in Tobin's Q , which is calculated from Compustat items as $(PRCC_F \times CSHO + AT - CEQ)/AT$. *State-level employee turnover rate* is the net job transitions (inflow minus outflow) as a percentage of total employment within the state. *Industry-level unionization rate* is the industry-level percentage of union members (from Unionstats.com). *State-level NCAs enforceability* is from Johnson et al. (2025). *Industry-level skill diversity* is calculated as one minus the Herfindahl-Hirschman Index (HHI) of skill concentration (measured by summing the squares of the employment shares at the skill level in each industry). *Industry-level skill supply* (from AI-exposed occupations) is measured as the employment-share-weighted average number of important skills required by AI-exposed occupations within the (2-digit SIC) industry. *Industry-level skill distance* is the employment-share-weighted average of the occupational skill distance across (each category) AI-exposed occupations. The between-occupation skill distance is proxied by the number of skills commonly required by each occupation pair (i.e., fewer common skills refer to a large skill distance). For each AI-exposed occupation, we calculate the average distance between that occupation and all others, where the distances are weighted by the number of job transitions to the focal occupation. *State-level UI benefits* is the state-level unemployment insurance benefits measured as the average weekly benefits times total weeks compensated. *Displaceable employment* is the size of a firm's total workforce that is in occupations exposed to displacing AI. *Displaceable wage* is calculated as a firm's yearly total wages paid to employees in occupations exposed to displacing AI.

Panel A: Occupation-by-firm characteristics

	N	Mean	SD	p10	Median	p90
Employment growth	761,232	1.330	89.178	-35.232	0.000	35.243
Employment growth (novel skills)	761,232	-0.043	0.381	0.000	0.000	0.000
Exposed to language AI	761,232	0.217	0.083	0.115	0.212	0.324
Exposed to perception AI	761,232	0.011	0.015	0.004	0.008	0.019
Exposed to motor control AI	761,232	0.915	0.059	0.85	0.924	0.972
Exposed to engagement AI	761,232	0.635	0.138	0.43	0.667	0.791
Exposed to decision-making AI	761,232	0.418	0.122	0.255	0.417	0.575
Exposed to learning AI	761,232	0.715	0.139	0.499	0.756	0.857
Exposed to creativity AI	761,232	0.902	0.078	0.818	0.917	0.973

(Continued)

Table 3, Continued

Panel B: Firm characteristics						
	N	Mean	SD	p10	Median	p90
<i>AI patenting per year</i>						
All types	18,779	12.101	93.115	0.000	0.000	12.000
Language	18,779	2.166	25.240	0.000	0.000	1.000
Perception	18,779	3.770	28.087	0.000	0.000	4.000
Motor control	18,779	0.892	5.984	0.000	0.000	1.000
Engagement	18,779	4.049	39.673	0.000	0.000	3.000
Decision-making	18,779	7.228	60.353	0.000	0.000	7.000
Learning	18,779	4.984	43.811	0.000	0.000	5.000
Creativity	18,779	1.278	14.674	0.000	0.000	1.000
Augmenting	18,779	8.784	72.877	0.000	0.000	8.000
Scope augmenting	18,779	6.450	56.649	0.000	0.000	6.000
Core augmenting	18,779	7.817	66.274	0.000	0.000	7.000
Displacing	18,779	4.094	29.530	0.000	0.000	4.000
<i>Other firm characteristics</i>						
Total assets (\$Millions)	18,507	18633.937	54029.415	46.412	1236.099	43437
ROA	18,464	-0.014	0.335	-0.395	0.094	0.207
Market-to-book	18,015	2.462	1.951	0.992	1.806	4.750
R&D (\$Millions)	16,096	435.549	1235.115	4.657	53.099	939.918
R&D Missing	18,507	0.13	0.337	0.000	0.000	1.000
TFP growth	15,750	-0.297	3.917	-2.394	-0.241	1.568
ΔSG&A (\$Millions)	14,804	0.056	0.181	-0.134	0.048	0.254
ΔOperating expense (\$Millions)	17,098	0.058	0.214	-0.167	0.051	0.298
ΔTobin's Q	16,574	-0.008	1.198	-0.97	0.012	0.894
State-level employee turnover rate	15,908	0.064	0.045	0.017	0.067	0.109
Industry-level unionization rate	18,499	7.992	3.042	4.093	7.850	11.258
State-level NCAs enforceability	16,093	0.601	0.32	0.065	0.734	0.904
Industry-level skill diversity	16,645	0.951	0.002	0.946	0.952	0.953
Industry-level skill supply	16,681	35.357	31.305	0.000	33.971	83.826
Industry-level skill distance	16,645	0.016	0.011	0.003	0.012	0.032
State-level UI benefits (\$Thousands)	14,282	809.447	673.49	114.579	614.089	1762.395
Displaceable employment (Thousands)	16,923	5.052	12.818	0.027	0.621	12.745
Displaceable wage (\$Millions)	16,923	336.751	825.349	2.746	47.918	837.537

Table 4. Effects of AI Innovation on Employment Growth

This table reports the results of regressions examining the effects of AI innovation on occupational employment growth. The sample consists of observations at the firm-by-occupation-by-year level. The dependent variable is the change in firm-by-occupation employment during a year divided by the firm's scaled total headcount in the prior year. In constructing this employment change variable, we divide a firm's total headcount by 10,000 for ease of exposition. The main explanatory variables are interactions involving *Exposed*—a binary variable equal to one if an occupation has at least four important skill requirements (O*NET importance score ≥ 3) that map to AI type k —and three different measures of firm-level patenting: (1) the natural log of (one plus) the year- t count of the firm's type- k patents; (2) the natural log of (one plus) the year- t count of the firm's other patents; and (3) the natural log of (one plus) the year- t overall count of the firm's patents. The regressions consider the seven different types of AI patents that correspond to the functional typology of AI outlined in Section 2. Other explanatory variables are described in Table 3 and measured as of the end of year $t-1$. Missing values of R&D are set to zero and captured with an indicator variable. Panel A presents baseline OLS regressions corresponding to Equation (1) in Section 4. Panel B presents second-stage 2SLS regression results corresponding to Equation (2) in Section 4. Instruments for the interaction variables are based on measures of patent examiner leniency as detailed in Section 4. Each regression includes firm fixed effects, (2-digit SIC) industry-by-year fixed effects, (3-digit SOC) occupation-by-year fixed effects, and fixed effects for the interaction of year and the firm's # of patent applications during the year. All continuous variables are winsorized at 1% and 99%. T-statistics, corresponding to standard errors clustered at the 2-digit SIC industry level, are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

Dependent Variable: Change in Firm-by-Occupation Employee Headcount							
Panel A: Baseline regressions (OLS)							
	Patent Type $k =$						
	Language AI (1)	Perception AI (2)	Motor control AI (3)	Engagement AI (4)	Decision- Making AI (5)	Learning AI (6)	Creativity AI (7)
Exposed $\times$ Log type- k patents	0.495 (1.23)	0.163 (0.51)	-0.534 (-1.21)	1.107*** (2.73)	0.873*** (3.43)	1.025** (2.54)	0.467 (1.42)
Exposed $\times$ Log other patents	3.528* (1.84)	4.153** (2.03)	3.620** (1.97)	2.614 (1.39)	2.697 (1.51)	2.893 (1.59)	3.718** (2.23)
[1-Exposed] $\times$ Log all patents	3.565* (1.92)	3.709** (2.03)	3.879** (2.03)	3.189* (1.71)	3.322* (1.85)	3.272** (2.05)	4.215** (2.49)
Log firm size	-3.825*** (-2.72)	-3.828*** (-2.72)	-3.827*** (-2.72)	-3.821*** (-2.70)	-3.822*** (-2.72)	-3.821*** (-2.72)	-3.820*** (-2.72)

(Continued)

Table 4, Continued

ROA	20.035*** (11.13)	20.03*** (11.12)	20.067*** (11.16)	20.069*** (11.18)	20.045*** (11.16)	20.038*** (11.19)	20.014*** (11.12)
Market-to-book	3.513*** (7.71)	3.511*** (7.70)	3.512*** (7.70)	3.516*** (7.69)	3.514*** (7.70)	3.516*** (7.68)	3.512*** (7.68)
Log R&D	-9.544*** (-3.95)	-9.545*** (-3.95)	-9.545*** (-3.95)	-9.544*** (-3.96)	-9.540*** (-3.96)	-9.549*** (-3.95)	-9.545*** (-3.95)
Fixed effects:							
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of applications $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.233	0.233	0.223	0.224	0.233	0.233	0.233
Observations	703,163	703,163	703,163	703,163	703,163	703,163	703,163

(Continued)

Table 4, Continued

Panel B: Instrumental variables regressions (2SLS, 2 nd stage)							
	Patent Type $k =$						
	Language AI (1)	Perception AI (2)	Motor control AI (3)	Engagement AI (4)	Decision- Making AI (5)	Learning AI (6)	Creativity AI (7)
Exposed $\times$ Log type- k patents	1.296*** (2.83)	-1.397*** (-2.44)	-1.866** (-2.04)	1.640*** (2.84)	1.397** (2.48)	1.979*** (2.73)	1.308** (2.41)
Exposed $\times$ Log other patents	-9.179 (-1.52)	-6.707 (-1.13)	-8.134 (-1.33)	-9.716 (-1.60)	-9.432 (-1.66)	-9.733* (-1.71)	-9.080 (-1.50)
[1-Exposed] $\times$ Log all patents	-8.407 (-1.38)	-8.462 (-1.39)	-8.210 (-1.32)	-8.352 (-1.36)	-8.273 (-1.40)	-8.330 (-1.40)	-7.648 (-1.24)
Log firm size	-3.638** (-2.53)	-3.643** (-2.53)	-3.646** (-2.53)	-3.621** (-2.50)	-3.662** (-2.55)	-3.649** (-2.53)	-3.637** (-2.51)
ROA	20.569*** (11.45)	20.556*** (11.45)	20.532*** (11.34)	20.552*** (11.53)	20.587*** (11.57)	20.550*** (11.41)	20.563*** (11.52)
Market-to-book	3.577*** (7.31)	3.575*** (7.31)	3.576*** (7.29)	3.578*** (7.30)	3.572*** (7.31)	3.566*** (7.26)	3.580*** (7.32)
Log R&D	-9.548*** (-4.01)	-9.554*** (-4.01)	-9.543*** (-4.00)	-9.534*** (-4.00)	-9.534*** (-4.00)	-9.506*** (-4.00)	-9.539*** (-4.01)
Fixed effects:							
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of applications $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F-statistic	14.371	14.533	14.120	14.254	16.301	14.447	14.891
Observations	703,163	703,163	703,163	703,163	703,163	703,163	703,163

Table 5. Effects of AI Innovation on Employment of Workers with Novel Skills

This table reports the results of regressions examining the effects of AI innovation on occupation-specific growth in the employment of workers with novel skills. The sample consists of observations at the firm-by-occupation-by-year level. The dependent variable is the change in firm-by-occupation employment of workers with novel skills, divided by the firm's scaled total headcount in the prior year. In constructing this change variable, we divide a firm's total headcount by 10,000 for ease of exposition. A worker has novel skills if he or she holds at least one skill that is not held by at least half of the workers in the same firm-by-occupation-by-year group. The main explanatory variables are interactions involving *Exposed*—a binary variable equal to one if an occupation has at least four important skill requirements (O*NET importance score ≥ 3) that map to AI type k —and three different measures of firm-level patenting: (1) the natural log of (one plus) the year- t count of the firm's type- k patents; (2) the natural log of (one plus) the year- t count of the firm's other patents; and (3) the natural log of (one plus) the year- t overall count of the firm's patents. The regressions consider the five types of AI patents that exhibit positive effects on employment growth in headcount, as reported in Table 4. Other explanatory variables are described in Table 3 and measured as of the end of year $t-1$. Missing values of R&D are set to zero and captured with an indicator variable. Panel A presents baseline OLS regressions corresponding to Equation (1) in Section 4. Panel B presents second-stage 2SLS regression results corresponding to Equation (3) in Section 4. Instruments for the interaction variables are based on measures of patent examiner leniency as detailed in Section 4. Each regression includes firm fixed effects, (2-digit SIC) industry by year fixed effects, (3-digit SOC) occupation by year fixed effects, and the firm's count of total patent applications by year fixed effects. All continuous variables are winsorized at 1% and 99%. T-statistics, corresponding to standard errors clustered at the 2-digit SIC industry level, are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

Dependent Variable: Change in Firm-by-Occupation Headcount of Employees with New Skills					
Panel A: Baseline regressions (OLS)					
	Patent Type k =				
	Language AI (1)	Engagement AI (2)	Decision-Making AI (3)	Learning AI (4)	Creativity AI (5)
Exposed $\times$ Log type- k patents	0.002 (0.66)	0.006*** (3.94)	0.002 (1.34)	0.005*** (3.25)	0.003* (1.90)
Exposed $\times$ Log other patents	-0.003* (-1.97)	-0.004*** (-3.20)	-0.002 (-1.03)	-0.003** (-2.29)	-0.002** (-2.02)
[1-Exposed] $\times$ Log all patents	0.0005 (0.40)	0.003** (2.07)	0.001 (0.84)	0.002 (1.54)	0.007*** (4.11)

(Continued)

Table 5, Continued

Log firm size	-0.0045*** (-3.40)	-0.004*** (-3.45)	-0.005*** (-3.49)	-0.004*** (-3.50)	-0.004*** (-3.31)
ROA	-0.010*** (-2.89)	-0.010*** (-2.92)	-0.010*** (-2.89)	-0.010*** (-2.94)	-0.010*** (-2.91)
Market-to-book	-0.00003 (-0.08)	-0.00002 (-0.09)	-0.00004 (-0.11)	-0.00003 (-0.11)	-0.00001 (-0.02)
Log R&D	-0.004** (-2.41)	-0.004** (-2.42)	-0.004** (-2.41)	-0.004** (-2.43)	-0.004** (-2.41)
Fixed effects:					
Firm	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes
Occupation $\times$ Year	Yes	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.148	0.149	0.148	0.148	0.150
Observations	703,163	703,163	703,163	703,163	703,163

(Continued)

Table 5, Continued

Panel B: Instrumental variables regressions (2SLS, 2 nd stage)					
	Patent Type $k =$				
	Language AI (1)	Engagement AI (2)	Decision-Making AI (3)	Learning AI (4)	Creativity AI (5)
Exposed $\times$ Log type-k patents	0.002 (0.88)	0.008*** (3.52)	0.004 (1.63)	0.011*** (3.64)	0.005*** (3.01)
Exposed $\times$ Log other patents	-0.030*** (-3.08)	-0.034*** (-3.38)	-0.031*** (-3.58)	-0.035*** (-3.76)	-0.031*** (-3.11)
[1-Exposed] $\times$ Log all patents	-0.027*** (-2.75)	-0.025** (-2.38)	-0.026*** (-2.84)	-0.025** (-2.48)	-0.020** (-2.02)
Log firm size	-0.004*** (-2.89)	-0.004*** (-2.80)	-0.004*** (-3.07)	-0.004*** (-3.03)	-0.004*** (-2.80)
ROA	-0.009*** (-2.75)	-0.009*** (-2.79)	-0.009*** (-2.75)	-0.009*** (-2.84)	-0.009*** (-2.80)
Market-to-Book	0.0001 (0.33)	0.0001 (0.36)	0.0001 (0.29)	0.0001 (0.23)	0.0001 (0.40)
Log R&D	-0.004** (-2.25)	-0.004** (-2.28)	-0.004** (-2.25)	-0.004** (-2.27)	-0.004** (-2.27)
Fixed effects:					
Firm	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes
Occupation $\times$ Year	Yes	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F statistic	14.371	14.254	16.301	14.447	14.891
Observations	703,163	703,163	703,163	703,163	703,163

Table 6. Effects of AI Innovation on Firm Productivity

This table reports the results of regressions examining the effects of labor-augmenting and labor-displacing AI innovation on firm-level productivity. The dependent variable is the growth rate of Total Factor Productivity (TFP) from year t to $t+1$. TFP is the estimated residual from a firm-year linear production function model in which firm-level production inputs are proxied for by property, plant, and equipment (PPENT) and total employment, and the dependent variable is the firm's total revenues. The main explanatory variables in Table 6 are the natural logarithms of (one plus) the year- t count of a firm's patents within a given category k and the year- t count of the firm's other patents. A patent is counted as of the date of its first office action. We consider four broad patent categories: *Augmenting*, *Scope augmenting*, *Core augmenting*, and *Displacing*. These categories consist of AI patent types that (according to Tables 4 and 5) affect exposed occupations by, respectively, increasing overall employment, increasing employment of workers with novel skills, increasing overall employment but not employment of employees with novel skills, and decreasing overall employment. Columns (1) through (4) report OLS regression results. Columns (5) through (8) report second-stage 2SLS regression results. Instruments for the patenting variables are based on measures of patent examiner leniency as detailed in Section 4. Other explanatory variables are described in Table 3 and measured as of the end of year t . Missing values of R&D are set to zero and captured with an indicator variable. Each regression includes firm fixed effects, (2-digit SIC) industry by year fixed effects, and the firm's count of total patent applications by year fixed effects. (see Equation (4) in Section 4 for more details). All continuous variables are winsorized at 1% and 99%. T-statistics, corresponding to standard errors clustered by 2-digit SIC industry, are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

Dependent Variable: Growth rate of Total Factor Productivity, Year t to Year $t+1$								
	Patent Category $k =$							
	Augmenting	Scope- Augmenting	Core- Augmenting	Displacing	Augmenting	Scope- Augmenting	Core- Augmenting	Displacing
	OLS				2SLS (2 nd Stage)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log category- k patents	0.045 (0.470)	-0.050 (0.410)	0.088 (1.100)	0.356** (2.543)	1.324*** (2.786)	1.036** (2.558)	1.162** (2.403)	0.303 (0.811)
Log other patents	-0.099 (0.431)	-0.072 (0.300)	-0.094 (0.396)	-0.051 (0.199)	1.938 (1.363)	2.212 (1.436)	1.933 (1.414)	2.669* (1.721)
Log firm size	-0.029 (0.273)	-0.029 (0.269)	-0.029 (0.278)	-0.029 (0.282)	-0.090 (0.862)	-0.082 (0.772)	-0.090 (0.860)	-0.088 (0.808)

(Continued)

Table 6, Continued

ROA	0.220*	0.216*	0.221*	0.216	0.202	0.171	0.193	0.154
	(1.702)	(1.678)	(1.701)	(1.652)	(1.546)	(1.350)	(1.506)	(1.242)
Market-to-book	-0.008	-0.008	-0.008	-0.008	-0.012	-0.011	-0.013	-0.014
	(0.462)	(0.466)	(0.467)	(0.458)	(0.651)	(0.621)	(0.696)	(0.722)
Log R&D	0.034	0.034	0.033	0.032	0.016	0.004	0.017	0.006
	(0.317)	(0.315)	(0.314)	(0.299)	(0.148)	(0.038)	(0.163)	(0.057)
Fixed effects:								
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.009	0.009	0.009	0.009				
Kleibergen-Paap Wald F					21.772	19.258	22.838	22.280
Observations	12,387	12,387	12,387	12,387	12,387	12,387	12,387	12,387

Table 7. Effects of Labor-Displacing AI Innovation on Cost Efficiency

This table reports the results of regressions examining the effects of labor-displacing AI innovation on innovating firms' cost efficiency. The dependent variable in columns (1) and (2) is the change from year t to year $t+1$ in the natural logarithm of (one plus) selling, general, and administrative expenditures (SG&A). In columns (3) and (4), the dependent variable is the change from year t to year $t+1$ in the natural logarithm of (one plus) operating expense. The main explanatory variables are *Log displacing patents* and *Log non-displacing patents*, the natural logarithms of (one plus) the year- t counts of a firm's patents that are displacing and non-displacing, respectively. A patent is counted as of the date of its first office action. Displacing patents are those AI patent types that, according to Table 4, affect exposed occupations by decreasing overall employment. Columns (1) and (3) report OLS regression results. Columns (2) and (4) report second-stage 2SLS regression results. Instruments for the patenting variables are based on measures of patent examiner leniency as detailed in Section 4. Other explanatory variables are described in Table 3 and measured as of the end of year t . Missing values of R&D are set to zero and captured with an indicator variable. Each regression includes firm fixed effects, (2-digit SIC) industry-by-year fixed effects, and fixed effects for the interaction between year and the firm's count of patent applications during the year. All continuous variables are winsorized at 1% and 99%. T-statistics, corresponding to standard errors clustered by 2-digit SIC industry, are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

Dependent Variable:	Δ SG&A		Δ Operating Expense	
	OLS	2SLS (2 nd Stage)	OLS	2SLS (2 nd Stage)
	(1)	(2)	(3)	(4)
Log displacing patents	0.001 (0.266)	-0.068*** (3.025)	0.010*** (2.851)	-0.040*** (2.712)
Log non-displacing patents	0.018*** (3.492)	-0.129** (2.288)	0.029*** (3.961)	-0.062 (1.211)
Log firm size	0.053*** (5.020)	0.056*** (5.732)	0.051*** (3.979)	0.052*** (4.285)
ROA	0.222*** (7.721)	0.222*** (7.586)	0.182*** (9.930)	0.185*** (9.892)
Market-to-book	0.028*** (13.572)	0.029*** (13.679)	0.032*** (10.139)	0.032*** (10.283)
Log R&D	-0.111*** (10.457)	-0.111*** (10.762)	-0.116*** (8.435)	-0.114*** (8.485)
Fixed effects:				
Firm	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes
Adjusted R ²	0.313		0.286	
Kleibergen-Paap rk Wald F-statistic		21.653		26.643
Observations	11,449	11,449	13,475	13,475

Table 8. Labor Augmentation, Labor Displacement, and Firm Value

This table reports the results of regressions examining the firm-value effects of labor-augmenting and labor-displacing AI innovation. The dependent variable is the change in Tobin's Q from year t to $t+1$. Tobin's Q is calculated from Compustat items as $(PRCC_F \times CSHO + AT - CEQ)/AT$. The main explanatory variables are the natural logarithms of (one plus) the year- t count of a firm's patents in a given category k and the year- t count of the firm's other patents. A patent is counted as of the date of its USPTO first office action. We consider four broad patent categories: *Augmenting*, *Scope Augmenting*, *Core Augmenting*, and *Displacing*. These categories consist of AI patents that (according to Tables 4 and 5) affect exposed occupations by, respectively, increasing overall employment, increasing employment of workers with novel skills, increasing overall employment but not employment of workers with novel skills, and decreasing overall employment. Columns (1) through (4) report OLS regression results. Columns (5) through (8) report second-stage 2SLS regression results. Instruments for the patenting variables are based on measures of patent examiner leniency as detailed in Section 4. Other explanatory variables are described in Table 3 and measured as of the end of year t . Missing values of R&D are set to zero and captured with an indicator variable. Each regression includes firm fixed effects, (2-digit SIC) industry-by-year fixed effects, and fixed effects for the interaction of year with the firm's total patent applications during the year. All continuous variables are winsorized at 1% and 99%. T-statistics, corresponding to standard errors clustered by 2-digit SIC industry, are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

Dependent Variable: Change in Tobin's Q, Year t to Year $t+1$								
	Patent Category $k =$							
	Augmenting	Scope- Augmenting	Core- Augmenting	Displacing	Augmenting	Scope- Augmenting	Core- Augmenting	Displacing
	OLS				2SLS (2 nd Stage)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log category- k patents	0.023 (0.947)	0.036 (1.512)	0.019 (0.756)	-0.002 (0.037)	0.347** (2.135)	0.335** (2.418)	0.321** (2.061)	0.414*** (2.944)
Log other patents	0.006 (0.175)	-0.007 (0.188)	0.014 (0.347)	0.035 (0.923)	0.330*** (2.737)	0.363** (2.528)	0.326*** (2.864)	0.314* (1.910)
Log firm size	-0.262*** (5.238)	-0.261*** (5.241)	-0.262*** (5.233)	-0.262*** (5.246)	-0.271*** (5.300)	-0.267*** (5.266)	-0.271*** (5.279)	-0.268*** (5.346)
ROA	0.048 (0.326)	0.048 (0.328)	0.047 (0.322)	0.048 (0.325)	0.044 (0.291)	0.035 (0.228)	0.042 (0.278)	0.037 (0.243)

(Continued)

Table 8, Continued

Market-to-book	-0.491*** (19.964)	-0.491*** (19.962)	-0.491*** (19.972)	-0.491*** (19.979)	-0.492*** (19.778)	-0.492*** (19.845)	-0.492*** (19.834)	-0.492*** (19.869)
Log R&D	0.016 (0.671)	0.016 (0.674)	0.016 (0.662)	0.016 (0.671)	0.012 (0.459)	0.009 (0.341)	0.011 (0.441)	0.011 (0.392)
Fixed effects:								
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# apps. $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.202	0.202	0.202	0.202				
Kleibergen-Paap Wald F					26.483	22.690	27.451	27.625
Observations	13,644	13,644	13,644	13,644	13,644	13,644	13,644	13,644

Table 9. Labor-Augmenting AI Innovation, Labor Market Frictions, and Firm Value

This table reports the results of Two-Stage Least Squares regressions examining the firm-value effects of augmenting AI innovation in subsamples of firms based on state-level or industry-level labor-market frictions. The dependent variable is the change in Tobin's Q from year t to $t+1$. The main explanatory variables are the natural logarithms of (one plus) the year- t count of a firm's patents within a given category k and the year- t count of the firm's other patents. A patent is counted as of the date of its USPTO first office action. We consider three broad patent types: *Augmenting*, *Scope Augmenting*, and *Core Augmenting*. These types consist of AI patents that (according to Tables 4 and 5) affect exposed occupations by, respectively, increasing overall employment, increasing employment of workers with novel skills, or increasing overall employment but not employment of workers with novel skills. Instruments for the patenting variables are based on measures of patent examiner leniency as detailed in Section 4. Panels A, B, and C measure labor-market frictions with, respectively, state-level employee turnover rates, industry-level labor unionization rates, and state-level enforceability of noncompete agreements (NCAs). A firm's headquarters state has a high (low) employee turnover rate if the net job transitions (inflow minus outflow) as a percentage of total employment within the state is above (below) the sample median. A firm's industry has a high (low) unionization rate if the percentage of industry-level union members as reported by *Unionstats.com* is above (below) the sample median. A firm's headquarters state has high (low) noncompete enforceability if the enforceability of NCAs is above (below) the sample median. Other explanatory variables are described in Table 3 and measured as of the end of year t . Missing values of R&D are set to zero and captured with an indicator variable. Each regression includes firm fixed effects, (2-digit SIC) industry-by-year fixed effects, and fixed effects for the interaction of year with the firm's count of patent applications during the year. All continuous variables are winsorized at 1% and 99%. T-statistics, corresponding to standard errors clustered by 2-digit SIC industry, are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

Panel A: State-level employee turnover rate

	Patent Category $k =$					
	Augmenting	Scope-Augmenting	Core-Augmenting	Augmenting	Scope-Augmenting	Core-Augmenting
	High In-State Turnover Rate			Low In-State Turnover Rate		
	(1)	(2)	(3)	(4)	(5)	(6)
Log category- k patents	0.443*** (2.735)	0.511** (2.280)	0.327* (2.011)	0.216 (1.098)	0.286 (1.316)	0.235 (1.200)
Log other patents	1.563*** (2.845)	1.591*** (3.028)	1.618*** (2.989)	0.673 (1.525)	0.738 (1.614)	0.627 (1.437)
Log firm size	-0.259** (2.464)	-0.259** (2.524)	-0.258** (2.468)	-0.288*** (5.888)	-0.289*** (5.845)	-0.288*** (5.682)

(Continued)

Table 9, Continued

ROA	-0.349 (1.209)	-0.347 (1.196)	-0.359 (1.245)	0.292* (1.690)	0.287 (1.575)	0.297* (1.725)
Market-to-book	-0.577*** (39.677)	-0.576*** (38.432)	-0.577*** (40.327)	-0.430*** (12.531)	-0.430*** (12.419)	-0.430*** (12.332)
Log R&D	-0.110 (0.899)	-0.112 (0.920)	-0.108 (0.871)	0.058 (0.877)	0.056 (0.845)	0.058 (0.894)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F	28.958	27.266	28.733	9.674	8.189	10.661
Observations	5,609	5,609	5,609	5,557	5,557	5,557

(Continued)

Table 9, Continued

Panel B: Industry-level unionization rate						
	Patent Category $k =$					
	Augmenting	Scope- Augmenting	Core- Augmenting	Augmenting	Scope- Augmenting	Core- Augmenting
	High Unionization Rate			Low Unionization Rate		
	(1)	(2)	(3)	(4)	(5)	(6)
Log category- k patents	0.024 (0.322)	0.219** (2.648)	-0.008 (0.084)	0.433* (1.830)	0.422** (2.499)	0.417* (1.761)
Log other patents	0.636** (2.463)	0.564** (2.075)	0.659** (2.500)	0.376 (0.742)	0.486 (0.766)	0.302 (0.644)
Log firm size	-0.271** (2.507)	-0.270** (2.511)	-0.271** (2.501)	-0.442*** (4.718)	-0.441*** (4.742)	-0.438*** (4.615)
ROA	-0.050 (0.196)	-0.042 (0.167)	-0.050 (0.194)	0.252** (2.374)	0.247** (2.222)	0.251** (2.419)
Market-to-book	-0.624*** (29.769)	-0.624*** (29.444)	-0.623*** (29.681)	-0.564*** (17.241)	-0.563*** (17.005)	-0.564*** (17.288)
Log R&D	-0.006 (0.136)	-0.011 (0.257)	-0.006 (0.123)	-0.026 (0.381)	-0.028 (0.409)	-0.026 (0.389)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F	24.042	23.289	22.005	29.549	18.325	32.557
Observations	6,663	6,663	6,663	5,772	5,772	5,772

(Continued)

Table 9, Continued

Panel C: State-level non-compete agreement (NCA) enforceability						
	Patent Category $k =$					
	Augmenting	Scope- Augmenting	Core- Augmenting	Augmenting	Scope- Augmenting	Core- Augmenting
	High NCA Enforceability			Low NCA Enforceability		
	(1)	(2)	(3)	(4)	(5)	(6)
Log category- k patents	0.570* (1.736)	0.744* (1.912)	0.455 (1.524)	0.425** (2.317)	0.350** (2.180)	0.403** (2.409)
Log other patents	0.849* (2.006)	0.750* (1.955)	0.808** (2.051)	0.052 (0.103)	0.097 (0.172)	0.045 (0.089)
Log firm size	-0.321*** (10.994)	-0.315*** (11.130)	-0.318*** (10.851)	-0.277** (2.535)	-0.271** (2.494)	-0.277** (2.555)
ROA	0.031 (0.199)	0.037 (0.250)	0.030 (0.193)	0.060 (0.247)	0.048 (0.196)	0.059 (0.248)
Market-to-book	-0.536*** (20.166)	-0.534*** (20.685)	-0.535*** (20.317)	-0.527*** (13.899)	-0.528*** (14.359)	-0.527*** (14.118)
Log R&D	-0.050 (0.992)	-0.047 (0.947)	-0.048 (0.988)	-0.011 (0.310)	-0.012 (0.333)	-0.012 (0.323)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
# apps $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F	71.713	90.119	77.697	12.734	9.777	13.088
Observations	6,646	6,646	6,646	5,727	5,727	5,727

Table 10. Labor-Augmenting AI Innovation, Labor Supply, and Firm Value

This table reports the results of Two-Stage Least Squares regressions examining the firm-value effects of augmenting AI innovation in subsamples of firms based on state-level or industry-level labor-market frictions. The dependent variable is the change in Tobin's Q from year t to $t+1$. The main explanatory variables are the natural logarithms of (one plus) the year- t count of a firm's patents within a given category k and the year- t count of the firm's other patents. A patent is counted as of the date of the USPTO first office action. We consider three broad patent types: *Augmenting*, *Scope Augmenting*, and *Core Augmenting*. These types consist of AI patents that (according to Tables 4 and 5) affect exposed occupations by, respectively, increasing overall employment, increasing employment of workers with novel skills, and increasing overall employment but not employment of workers with novel skills. Instruments for the patenting variables are based on measures of patent examiner leniency as detailed in Section 4. In Panel A, industry-level *skill diversity* is calculated as one minus the Herfindahl-Hirschman Index (HHI) of skill concentration (measured by summing the squares of the employment shares at the skill level in each industry). In Panel B, we measure the industry-level *skill supply from AI-exposed occupations* as the employment-share-weighted average number of important skills required by AI-exposed occupations within the (2-digit SIC) industry. In Panel C, we measure the industry-level *skill distance* as the employment-share-weighted average of occupational skill distances among AI-exposed occupations within an industry. For each AI-exposed occupation, we compute its average skill distance to all other occupations, weighting each pair by the volume of job transitions into the focal occupation. The skill distance between two occupations is measured by their overlap in required skills, with fewer shared skills indicating greater distance. Other explanatory variables are described in Table 3 and measured as of the end of year t . Missing values of R&D are set to zero and captured with an indicator variable. Each regression includes firm fixed effects, (2-digit SIC) industry-by-year fixed effects, and fixed effects for the interaction of year with the firm's count of patent applications during the year. All continuous variables are winsorized at 1% and 99%. T-statistics, corresponding to standard errors clustered by 2-digit SIC industry, are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

Panel A: Industry-level skill diversity						
	Patent Category k =					
	Augmenting	Scope-Augmenting	Core-Augmenting	Augmenting	Scope-Augmenting	Core-Augmenting
	High Skill Diversity			Low Skill Diversity		
	(1)	(2)	(3)	(4)	(5)	(6)
Log category- k patents	0.828*** (2.770)	1.072*** (4.024)	0.742** (2.486)	-0.178 (1.756)	-0.108 (1.127)	-0.157 (1.494)
Log other patents	0.126 (0.341)	0.359 (1.006)	0.154 (0.441)	-0.640 (1.626)	-0.641 (1.478)	-0.647* (1.781)
Log firm size	-0.248*** (4.600)	-0.246*** (4.355)	-0.248*** (4.700)	-0.256** (2.688)	-0.256** (2.625)	-0.255** (2.696)

(Continued)

Table 10, Continued

ROA	-0.093 (1.616)	-0.119** (2.069)	-0.099* (1.776)	0.219 (0.383)	0.224 (0.393)	0.227 (0.396)
Market-to-book	-0.546*** (100.617)	-0.545*** (102.195)	-0.546*** (103.830)	-0.478*** (9.557)	-0.478*** (9.579)	-0.478*** (9.575)
Log R&D	-0.013 (0.630)	-0.023 (0.928)	-0.017 (0.826)	0.012 (0.263)	0.014 (0.313)	0.011 (0.234)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F	23.580	18.345	21.164	17.298	18.308	15.459
Observations	6,006	6,006	6,006	5,942	5,942	5,942

(Continued)

Table 10, Continued

Panel B: Industry-level skill supply from AI-exposed occupations						
	Patent Category $k =$					
	Augmenting	Scope- Augmenting	Core- Augmenting	Augmenting	Scope- Augmenting	Core- Augmenting
	High Skill Supply			Low Skill Supply		
	(1)	(2)	(3)	(4)	(5)	(6)
Log category- k patents	0.529** (2.429)	0.628*** (4.004)	1.154*** (3.165)	0.042 (0.361)	0.064 (0.569)	-0.135 (0.880)
Log other patents	1.076*** (5.961)	1.148*** (4.499)	0.162 (0.396)	-0.003 (0.007)	-0.014 (0.032)	-0.498 (0.944)
Log firm size	-0.258*** (2.956)	-0.251*** (2.893)	-0.249*** (5.825)	-0.225*** (3.409)	-0.225*** (3.469)	-0.256*** (3.232)
ROA	-0.101 (0.351)	-0.112 (0.389)	-0.038 (0.335)	-0.029 (0.134)	-0.028 (0.133)	-0.029 (0.064)
Market-to-book	-0.508*** (7.871)	-0.509*** (7.942)	-0.535*** (37.563)	-0.544*** (24.609)	-0.544*** (24.747)	-0.464*** (10.833)
Log R&D	-0.056 (0.665)	-0.061 (0.741)	-0.010 (0.409)	-0.050* (2.001)	-0.051* (2.028)	0.003 (0.098)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F	24.878	14.218	22.083	27.614	39.469	13.084
Observations	5,962	5,962	6,087	5,849	5,849	5,827

(Continued)

Table 10, Continued

Panel C: Industry-level skill distance between AI-exposed occupations and others						
	Patent Category $k =$					
	Augmenting	Scope- Augmenting	Core- Augmenting	Augmenting	Scope- Augmenting	Core- Augmenting
	Large Skill Distance			Small Skill Distance		
	(1)	(2)	(3)	(4)	(5)	(6)
Log category- k patents	0.465** (2.631)	0.425*** (2.731)	-0.165 (0.870)	0.022 (0.090)	0.421 (1.726)	0.735** (2.128)
Log other patents	0.269 (1.054)	0.367 (1.203)	-0.372 (0.561)	0.718 (0.937)	0.632 (0.765)	0.393 (1.063)
Log firm size	-0.263*** (3.395)	-0.257*** (3.301)	-0.264*** (4.038)	-0.236*** (3.116)	-0.222*** (2.883)	-0.247*** (4.959)
ROA	-0.060 (0.319)	-0.075 (0.381)	0.011 (0.026)	0.035 (0.079)	0.043 (0.098)	-0.048 (0.311)
Market-to-book	-0.542*** (42.175)	-0.542*** (42.253)	-0.459*** (10.914)	-0.446*** (12.198)	-0.447*** (12.261)	-0.540*** (45.720)
Log R&D	-0.021 (1.046)	-0.028 (1.198)	-0.026 (1.045)	0.010 (0.159)	0.010 (0.151)	-0.013 (0.413)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F	24.712	25.899	23.834	43.266	28.874	17.012
Observations	5,967	5,967	5,954	5,553	5,553	5,632

Table 11. Labor-Displacing AI Innovation, Potential Cost Savings, and Firm Value

This table reports the results of Two-Stage Least Squares regressions examining the firm-value effects of displacing AI innovation in subsamples of firms with different degrees of potential cost savings. The dependent variable is the change in Tobin's Q from year t to year $t+1$. The main explanatory variables are *Log displacing patents* and *Log non-displacing patents*, the natural logarithms of (one plus) the year- t counts of a firm's patents that are displacing and non-displacing, respectively. A patent is counted as of the USPTO first office action date. Displacing patents are those AI patents that, according to Table 4, affect exposed occupations by decreasing overall employment. Instruments for the patenting variables are based on measures of patent examiner leniency as detailed in Section 4. A firm's headquarters state has a high (low) level of *unemployment insurance (UI) benefits* if the state's average weekly benefits times total weeks compensated is above (below) the sample median. A firm has high (low) *Displaceable employment* if a firm's total workforce who are in occupations exposed to displacing AI is above (below) the sample median. A firm has high (low) *Displaceable wage* if a firm's total annual wages paid to employees in occupations exposed to displacing AI is above (below) the sample median. Other explanatory variables are described in Table 3 and measured as of the end of year t . Missing values of R&D are set to zero and captured with an indicator variable. Each regression includes firm fixed effects, (2-digit SIC) industry-by-year fixed effects, and fixed effects for the interaction of year with the firm's total patent applications during the year. All continuous variables are winsorized at 1% and 99%. T-statistics, corresponding to standard errors clustered by 2-digit SIC industry, are reported in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

Dependent Variable: Change in Tobin's Q, Year t to Year $t+1$						
	UI benefits		Displaceable Employment		Displaceable Wage	
	High	Low	High	Low	High	Low
	(1)	(2)	(3)	(4)	(5)	(6)
Log displacing patents	0.312 (1.682)	0.644*** (3.513)	0.227* (1.910)	-0.282 (0.850)	0.346** (2.177)	-0.121 (0.326)
Log non-displacing patents	0.244 (0.346)	0.442 (0.692)	-0.327 (0.589)	0.587 (1.416)	-0.061 (0.132)	0.363 (1.008)
Log firm size	-0.285*** (3.244)	-0.266*** (6.940)	-0.229** (2.514)	-0.304*** (5.432)	-0.238*** (2.751)	-0.293*** (4.782)
ROA	-0.092 (0.488)	0.174 (1.267)	0.297 (0.471)	-0.005 (0.031)	0.494 (0.976)	-0.026 (0.186)
Market-to-book	-0.545*** (52.725)	-0.526*** (16.704)	-0.411*** (14.371)	-0.531*** (22.298)	-0.432*** (18.202)	-0.541*** (25.949)

(Continued)

Table 11, Continued

Log R&D	0.004 (0.039)	-0.001 (0.027)	-0.028 (1.042)	0.022 (0.387)	-0.060* (1.965)	0.018 (0.349)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
# applications $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F statistic	77.518	25.970	14.129	30.335	15.280	38.157
Observations	4,707	5,190	6,143	6,272	6,119	6,289